

# **Wheels to Meals: Measuring the Impact of Micromobility on Restaurant Demand**

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## **Abstract**

Dockless shared micromobility services have grown substantially in recent years, but their impact upon consumer demand has remained largely unstudied. The authors estimate how the largest and fastest growing segment of this market – the dockless electric scooter (‘e-scooter’) sharing industry – impacts spending in one of the largest segments of the local economy, the restaurant industry. Using data covering 391 companies in 98 U.S. cities, the authors find that the introduction of e-scooters in a city significantly impacts restaurant spending, increasing spending by approximately 5.2% for e-scooter users, driving incremental spending of at least \$11.3 million annually across all cities that first allowed e-scooters to operate over summer 2018. Impact varies by restaurant subcategory, with a stronger positive effect upon fast food restaurant spending, and a weaker effect upon sitdown restaurant spending. E-scooter entry has a larger impact upon companies with higher historical revenues selling at lower prices. It facilitates discovery of new restaurants from prospective customers and repeat business from already-acquired customers.

**Keywords:** micromobility, sharing economy, electric scooters

Dockless shared micromobility has expanded significantly since its inception in September 2017. It is a shared transportation service in which companies make low-speed vehicles – electric scooters (‘e-scooters’) and electric bicycles – available to consumers in any approved location, and enable consumers to find and rent them on a short-term basis through mobile apps. We focus upon the e-scooter industry, which is the largest and fastest-growing subsegment of the dockless shared mobility industry. Over 100 thousand e-scooters were deployed in the United States in 2019<sup>1</sup> and many more in the rest of the world.<sup>2</sup> E-scooters have become a popular form of last-mile transportation due to their convenience and ease of use for consumers, and their ability to help cities meet mobility, climate/emissions and equity goals with a limited need for dedicated infrastructure (City of Portland 2019, City of Washington D.C. 2018).

The research questions we study in this paper revolve around the impact of e-scooters on consumer demand in the restaurant category and what this implies for the local economy:

1. When a city allows e-scooters to operate in it, what effect does this have upon spending in the restaurant category from e-scooter users?
2. What types of cities, and what types of businesses, are most affected?
3. Are these effects offset by changes in local non-restaurant spending, or do they carry through to overall spending in the local economy?
4. How large is restaurant spending lift relative to the cost to a city of e-scooter rollout?

Our intended contribution in this article is to formally evaluate these questions through an individual-level analysis of consumer purchase behavior, leveraging a unique credit card panel data set from Earnest Analytics, a leading data analytics company. Through this data set, we observe individual-level purchase behavior for 2.2 million individuals at all 391 companies in its coverage universe within the restaurant sector from January 2017 through February 2020 (referred to hereafter as the ‘observation period’).

It is not obvious, *a priori*, that we would observe significant effects. Cities did not launch e-scooter programs to support their local economies. Indeed, a meta-analysis of publicly disclosed reports produced by cities about their e-scooter programs shows that the primary

purposes for launching these programs were reducing traffic congestion and reducing carbon emissions. Cities do not expect meaningful restaurant spending effects, and e-scooter operators have not been marketing financial spillovers to cities as part of their proposals. As such, uncovering that e-scooter program introduction does have significant effects for e-scooter users, and showing that those effects flow through into meaningful changes in overall spending in the local economy would have unexpected yet meaningful consequences for city legislators (for whom this may influence local economic growth and sales tax revenue) and e-scooter operators (for whom this is a positive externality that they could further capitalize upon).

We analyze transaction data for 46,451 individuals residing in 98 cities, 49 of which we designate as “treatment cities,” with the remainder representing “control cities.” Treatment cities collectively represent every city that first allowed e-scooters to operate over the five-month period between June 2018 and October 2018, roughly corresponding to summer 2018, which was the first full summer after the inception of the e-scooter industry. Over 50% of all cities that ever allowed e-scooters to operate, from their inception in September 2017 through February 2020, did so over this one summer period. Control cities are similar to the treatment cities in terms of observable sociodemographic variables, but did not allow e-scooters to operate over the observation period.

Our main analysis compares individual-level spending in treatment and control cities using a fixed effects regression model and coarsened exact matching to account for observed and unobserved city- and individual-level confounders. We also run a number of robustness checks including placebo tests and other analyses to assess the robustness of our results to a variety of forms of endogeneity.

Our main result is that introducing an e-scooter program into a city generates significant positive economic spillover from e-scooter users in the restaurant sector, increasing restaurant spending for these individuals by 5.2%. This translates into an average spending uplift over a one-year period of approximately \$179.1 per e-scooter allowed across the subset of cities we study. However, analysis of heterogeneity in these effects shows significant variation as a function of business and city characteristics. For example, we observe a

larger positive effect upon spending at fast food restaurants such as McDonald's, and a much smaller effect upon sitdown restaurants such as Olive Garden, implying that all restaurants do not benefit from e-scooter entry to the same extent. We do not find evidence that the increase in spending in the restaurant category is coming at the expense of local businesses outside of the restaurant category, implying that e-scooter program introduction could be an overall net positive for the local economy.

An examination of the reasons why individuals ride e-scooters supports our results. We collected all publicly accessible reports, to the best of our knowledge, that were conducted by cities about their e-scooter pilot programs. 20 cities produced such surveys, 15 of whom surveyed riders about trip purpose. Trip purpose data from 7 of these cities is depicted in Figure 1. While there was large variation in the framing of the survey questions, they consistently reported that individuals ride e-scooters to ride to or from restaurants. Figure 1 shows, for example, that 42% of all e-scooter trips in Chicago were to or from a restaurant (City of Chicago 2020), while in Seattle, Santa Monica and Milwaukee, 15%, 15.5% and 14.4% of respondents stated that the primary purpose of their last trip was dining (City of Milwaukee 2019, City of Santa Monica 2019, City of Seattle 2022). We readily acknowledge that many of these restaurant visits would likely still have occurred even if e-scooters were not available. E-scooters did not cause all of these restaurant visits. That being said, the high proportion of e-scooter trips resulting in restaurant visits is nevertheless suggestive that introducing an e-scooter program may affect restaurant spending.<sup>3</sup>

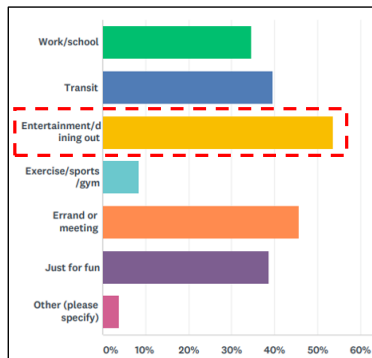
These surveys report that unlike other forms of shared mobility such as ridehailing, individuals typically ride e-scooters for hedonic purposes. A survey conducted in Portland showed that 58% and 53% of respondents listed fun / recreation and social / entertainment as one of their three most common uses, respectively (City of Portland 2020), with similar results in other cities. Taken together, these findings may be suggestive of a mechanism akin to individuals enjoying an unplanned trip to have a quick meal or drink with friends.

Our paper joins a growing literature studying the effects of growth in the sharing economy, with Airbnb being the company most studied. Zervas, Proserpio, and Byers (2017) study Airbnb's entry into the state of Texas and quantify its economic impact on the hotel

Figure 1: E-scooter trip usage survey results for selected cities

**Alexandria VA**

Q4 What kinds of trips do you take on e-scooters?

**Seattle WA**

What was the main purpose of your LAST scooter share trip?

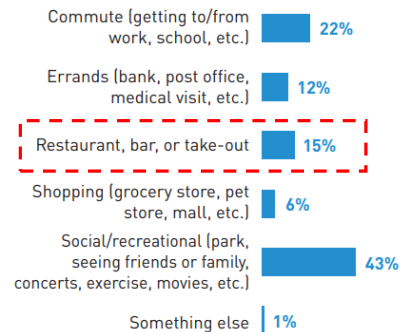
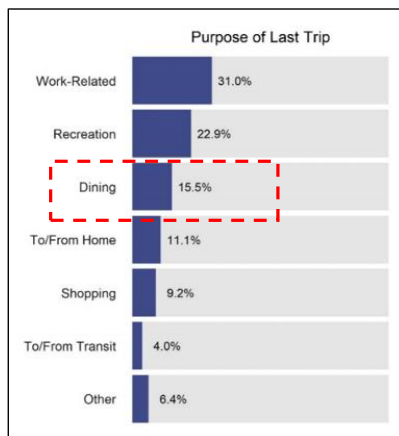
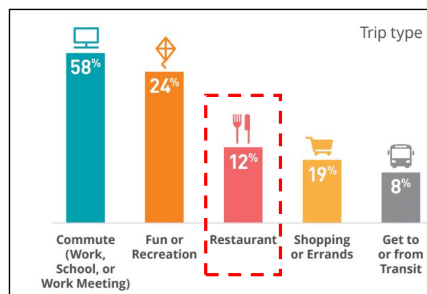
**Chicago IL**

Table 7: Trip Purpose for E-scooter Use\*

|                                   | Commute | Go To or From Transit | Go To or From School | Social/ Entertainment | Go To or From Restaurant |
|-----------------------------------|---------|-----------------------|----------------------|-----------------------|--------------------------|
| More than 1x per day              | 75%     | 49%                   | 10%                  | 32%                   | 30%                      |
| Daily                             | 70%     | 46%                   | 11%                  | 35%                   | 31%                      |
| 3-6x per week                     | 60%     | 50%                   | 6%                   | 41%                   | 40%                      |
| 1-3x per week                     | 38%     | 43%                   | 7%                   | 50%                   | 47%                      |
| Occasionally, but < once per week | 23%     | 32%                   | 3%                   | 58%                   | 48%                      |
| I've only ridden once             | 10%     | 12%                   | 1%                   | 39%                   | 21%                      |
| Overall                           | 30%     | 34%                   | 4%                   | 50%                   | 42%                      |
| Visitors                          | N/A     | 36%                   | N/A                  | 41%                   | 33%                      |

**Santa Monica CA****Portland OR****Milwaukee WI**

| What is the most frequent reason you've ridden a dockless scooter? | 1 trip | 2+ trips | All   |
|--------------------------------------------------------------------|--------|----------|-------|
| Connecting to transit (bus/streetcar)                              | 2.1%   | 1.7%     | 1.6%  |
| Riding for fun or recreation                                       | 50.1%  | 24.4%    | 28.6% |
| Running errands or shopping                                        | 2.5%   | 5.0%     | 4.6%  |
| Traveling to/from a restaurant                                     | 11.2%  | 15.0%    | 14.4% |
| Traveling to/from a work-related meeting or appointment            | 4.4%   | 6.6%     | 6.2%  |
| Traveling to/from entertainment                                    | 19.3%  | 26.4%    | 25.2% |
| Traveling to/from school or campus                                 | 1.7%   | 3.4%     | 3.6%  |
| Traveling to/from work                                             | 2.9%   | 15.4%    | 13.4% |
| Other                                                              | 5.8%   | 1.6%     | 2.3%  |

**San Antonio TX**

How did you use e-scooters/e-bikes? (Select all that apply)

| ANSWER CHOICES                   | RESPONSES |
|----------------------------------|-----------|
| To get to a shopping destination | 40.73%    |
| To get to a restaurant           | 56.95%    |
| For sightseeing                  | 70.20%    |
| For recreation                   | 58.61%    |
| For transportation               | 69.21%    |
| Total Respondents: 302           |           |

*Note:* Figures and tables are obtained from the following survey reports: City of Alexandria (2019), City of Seattle (2022), City of Chicago (2020), City of Santa Monica (2019), City of Portland (2020), City of Milwaukee (2019), City of San Antonio (2019). Red boxes overlaid by the authors highlight data specific to the restaurant category.

industry. Basuroy, Y. Kim, and Proserpio (2022) study Airbnb's effect upon the restaurant industry, while Barron, Kung, and Proserpio (2021) analyze its effect upon housing affordability. These papers are similar to ours in that they seek to quantify Airbnb's impact upon a financially oriented measure of interest (e.g., hotel or restaurant revenue, home prices, and housing rental rates).

The rest of the paper is organized as follows. In the next section, we summarize the data we use to perform the analysis. We then develop our model and discuss its identification, present the main empirical results, and outline the robustness checks that we performed before concluding with policy implications and a discussion of the results and future work.

## *DATA*

Our main data source is an individual-level panel data set from Earnest Analytics, a leading data analytics firm that has access to de-identified credit and debit card transactions at the daily level. We obtain purchase data for all 391 companies tracked by the data provider in the restaurant category for our main analysis. Each company is further subcategorized as a fast food restaurant (222 companies) or a sitdown restaurant (169). Table 1 contains summary statistics of this data. It shows, for example, that 78.8% of the individuals in the data set visited a restaurant at least once over the observation period, that there are an average of 2.17 purchase-weeks at a restaurant per individual (hereafter referred to interchangeably as 'panel member') in a given month, and that individuals spend \$30.95 at restaurants each purchase-week, on average. The restaurant category only represents in-store spending, as this is the category of spending that we would expect a priori to be impacted by e-scooter introduction. The data set includes an indicator variable for whether a purchase was made in-store, which we use to exclude purchases that were not made in-store.

We also obtain transaction data at companies in other categories – restaurant delivery (31 companies), online grocers (35), air travel (13), mass transit (16), gas station (47), ridesharing (5), events and attractions (112), and home entertainment (246) – which we use as matching variables and for placebo tests. In Web Appendix A, we list all companies in the restaurant category, as well as all companies in these other non-restaurant categories.

Table 1: Summary Statistics for Food Spending per Customer by Category

| Category          | Active Panel Members |                | Average Purchase-Week Frequency per Month |         | Spend per Purchase-Week (\$) |         |
|-------------------|----------------------|----------------|-------------------------------------------|---------|------------------------------|---------|
|                   | Count                | Share of Panel | Mean                                      | St. Dev | Mean                         | St. Dev |
| Total Restaurants | 76,463               | 78.8%          | 2.167                                     | 0.429   | 30.95                        | 26.2    |
| Fast food         | 74,232               | 76.5%          | 2.047                                     | 0.410   | 23.75                        | 22.61   |
| Sitdown           | 30,567               | 31.5%          | 0.453                                     | 0.09    | 40.12                        | 21.9    |

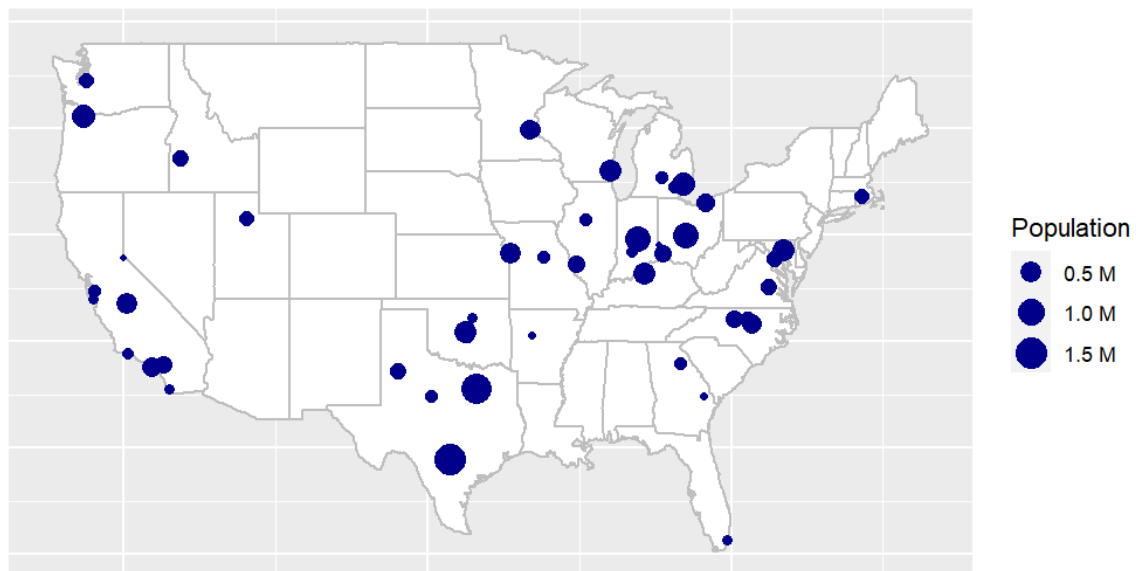
*Note:* Active panel members indicates people who made a purchase at least once during the observation period. Count is the number of active panel members, and share of panel represents the proportion of active panel members out of the full panel. Average purchase-week frequency per month is the total number of purchase-weeks per member-month, averaging across all member-months. Spend per purchase-week is spend amount per week, averaging across all purchase-weeks. All statistics are computed over the full observation period.

Through this data set, we also observe the location at the city-level associated with most in-store transactions. We use this location data to subset down to individuals making the majority of their purchases in 98 cities from January 1, 2017 through February 29, 2020. As mentioned in the previous section, these 98 cities consist of 49 “treatment cities,” representing all cities that launched e-scooter programs over summer 2018,<sup>4</sup> which we match against 49 “control cities” that are similar to our treatment cities in terms of sociodemographics, but did not launch e-scooter programs over the observation period. A table containing all treatment and control cities is available in Web Appendix B.

Cities launching e-scooter programs over this period vary significantly in terms of observable 2018 demographics, including population (e.g., South Lake Tahoe, California and San Antonio, Texas have populations of 22 thousand and 1.5 million, respectively), income (e.g., Oxford, Ohio and Arlington, Virginia have median household incomes of \$27 thousand and \$120 thousand), and climate (e.g., Saint Paul, Minnesota and Coral Gables, Florida have average annual temperatures of 44 and 76, respectively). These cities also vary significantly in terms of their geographical location. Through Figure 2, which plots the location of all cities launching e-scooter programs over summer 2018, it is visually evident that these cities (and their population sizes) are broadly distributed across the country. This variability enables us to identify heterogeneous treatment effects as a function of city characteristics, expanding the relevance of our analysis to policymakers in other cities looking to understand what effect launching an e-scooter program may have in their city, given their city’s

observable demographics.

Figure 2: Map of Cities Launching E-scooter Programs



*Note:* Dots represent 49 cities launching e-scooter programs from June 2018 to October 2018. The area of each dot is proportional to the population of the associated city.

Our unit of analysis is an individual over a weekly unit of time within a city. Although we could in theory marginalize our data across customers and perform an aggregate analysis as in Zervas, Proserpio, and Byers (2017), we proceed at the individual-level instead for two reasons. First, even though e-scooters have become more popular in recent years, they are still a small proportion of overall transportation. As such, the effects that we would expect to see at the aggregate-level would be obfuscated by other sources of variation that are unrelated to the phenomenon that we are studying. Second, an individual-level analysis provides us with greater visibility into the effects that an e-scooter program has, and how those effects vary across customers. Our geographical unit is at the city-level because our transaction data is only available at the city-level. A city-level analysis is most relevant to policymakers, as policymakers decide whether to launch an e-scooter program at the city-level, and overall employment and tax revenue are driven off of city-level spending. Also there are additional endogeneity concerns stemming from possible dynamics within subregions of a city – as in Bertrand, Duflo, and Mullainathan (2004), and collapsing the data to a higher level of aggregation allays such concerns.

The observation period for our main analysis is comprised of a pre-treatment period,

over which both treatment and control cities do not have e-scooter programs, and a post-treatment period, over which treatment cities have e-scooter programs but control cities do not. Our observation period is 38 months long, from January 2017 to February 2020 (before the onset of the COVID-19 pandemic).

We pre-process the data, filtering down to individuals living within the cities of interest, dropping transactions generated outside of home cities (e.g., vacation-related spending), and removing inactive accounts. For preprocessing details, see Web Appendix C.

*Representativeness.* While the panel data set is very large (representing approximately 4.1% of the working age and older population in the cities we study), there may still be concerns regarding the representativeness of the patterns in restaurant spending of our panel members relative to those in the broader population. We empirically assess the external validity of our panel data by obtaining population-level data for all publicly-traded, US-listed restaurant brands that disclose sales measures. 36 restaurant brands met these criteria (e.g., Burger King, Domino's Pizza, and McDonald's), collectively representing 55.4% of total panel sales in our 391-company coverage universe, providing us with "gold standard" population-level spending data. We obtain the empirical correlation between these sales disclosures and the corresponding panel-level sales figures over all available quarterly observations from the first calendar quarter of 2018 through the third calendar quarter of 2021. The sample average and median of these 36 correlations were 91.8% and 98.6%, respectively, and the interquartile range was 92.3% to 99.1%. While not perfect, these statistics are nevertheless high, supporting the notion that restaurant spending patterns as observed through our panel are reflective of population-level trends. For a list of companies, panel-population correlations, and a more detailed discussion of the representativeness analysis, see Web Appendix D.

## *IDENTIFICATION AND MODEL DEVELOPMENT*

### *Empirical Strategy*

While consumer adoption of e-scooters is growing, the category is still young. A small but growing proportion of individuals in treated cities – 15% as of February 2020 – had used

an e-scooter at least one time. We would not expect the introduction of an e-scooter program to impact restaurant spending for individuals who do not ride e-scooters, but because non-riders represent a large proportion of the population as of the end of the observation period, variation in their spending would obscure our estimate of the impact of e-scooter program introduction from those who do ride (which we would expect, a priori, may be economically meaningful but not large in magnitude). Therefore, we focus upon e-scooter users – individuals who ride at least once during the observation period – as the treated individuals in our analysis. Our estimate can be interpreted as the effect of the treatment on compliers, where treatment is assigned when a city introduces an e-scooter program, and compliers are e-scooter users. However, these individuals may be systematically different from individuals who are not e-scooters users, creating potential observed and unobserved confounders which, if not controlled for, could bias our results. In this section, we describe how we control for these and other potential confounders to satisfy the unconfoundedness assumption.

Confounding may arise at the individual-level, as described above, and at the city-level, through cities' decisions to introduce e-scooter programs. Confounders for both sources may be observable, unobservable and static, or unobservable and time-varying. Therefore, our empirical strategy must control for all of the resulting forms of confounding bias. We combine matching – treatment cities to control cities, and treated individuals to control individuals – to control for observable sources of confounding from both sources. We further control for static unobservable sources of confounding at both the city- and individual-level through a difference-in-differences (DID) approach with fixed effects. We assess the robustness of our results to various manifestations of time-varying unobservable confounders through a collection of supplementary analyses, including a qualitative analysis of the city-level decision to introduce e-scooter programs, parallel pre-treatment trends, placebo tests, a permutation test, and a variety of other robustness checks. We discuss matching, the two-way fixed effects DID model specification, the e-scooter entry decision, and parallel pretrends next, leaving the other robustness checks for after we present the main results in the Empirical Results section.

### *Individual-Level Matching*

To control for observable individual-level sources of confounding, we use matching to identify individuals in control cities who are as similar as possible to treated individuals in terms of their propensity to be e-scooter users. Matching provides us with a more credible estimate of the counterfactual spending baseline for treated individuals. It also lowers model dependence and improves estimation efficiency and stability, improving the credibility of causal inferences (Imbens 2014).

The e-scooter survey results mentioned in the Introduction section inform us of e-scooter trip purpose and what mode of transit those trips replaced (e.g., to go to an event, or to replace a ridesharing trip). We leverage this information to construct a collection of matching variables representing spending in the following categories: events and attractions, gas station, home entertainment, mass transit, restaurant delivery, and ridesharing. These categories span the most popular destinations for e-scooter trips, popular substitutes (e.g., home entertainment), as well as mode shifted categories or proxies thereof (e.g., ridesharing and gas station). Finally, we also include total spend in the panel as a matching variable, which serves as a proxy for income. There were 13,583 and 32,868 treated and control units with strictly positive weights after matching, with 0.5% and 56.8% of treated and control units dropped due to the matching procedure. Performing this procedure significantly improved pre-treatment covariate balance, with a 55% reduction in standardized mean difference across matching covariates, on average. For details of this procedure, see Web Appendix E.

While our focal matching method is coarsened exact matching (CEM) because of its well-documented advantages vis a vis more traditional matching methods such as propensity score matching (PSM), our conclusions do not hinge upon using CEM. As we show in Table 3, results are consistent in magnitude and direction when we use PSM, and when use no matching at all.

### *City-Level Matching*

We similarly control for observable sources of variation that may affect the propensity of a city to introduce an e-scooter program through matching. We do so by performing optimal full matching (Hansen and Klopfer, 2006) using a collection of matching variables that include population, population density, average annual temperature, average annual precipitation, and median household income. Web Appendix B and E provide implementation details and the resulting list of control cities.

### *Difference-in-Differences*

We apply a generalized DID model with individual and time fixed effects to control for static unobserved city- and individual-level confounders, and to further control for common demand shocks and city-specific spending trends that might affect restaurant spending and e-scooter entry.

Letting  $\log(\text{Spend}_{it} + 1)$  represent the log of the total amount spent (plus one) for each consumer  $i$  in week  $t$ , and letting  $\text{ScooterEntry}_{c(i)t}$  denote a binary variable equal to 1 if an e-scooter program had already begun in city  $c$  by week  $t$  and 0 otherwise, we assume the following:

$$(1) \quad \log(\text{Spend}_{it} + 1) = \beta_0 + \beta_1 \text{ScooterEntry}_{c(i)t} + \phi_i + \xi_t + \theta_c t + \epsilon_{it},$$

where  $\beta_1$  is the coefficient of interest, measuring how e-scooter program introduction affects (log) spending in the restaurant category for e-scooter users,  $\phi_i$  and  $\xi_t$  represent individual-specific fixed effects and week fixed effects respectively,  $\theta_c t$  is a city-specific linear time trend, and  $\epsilon_{it}$  is an error term.

$\phi_i$  control for time-invariant individual-specific characteristics such as static but unobserved preferences, typical transit patterns, and sociodemographic factors such as gender, age, and educational background, as well as city-specific characteristics such as population density, walkability, public transit availability, the size of the downtown area, and other factors that effectively do not change over the observation period.  $\xi_t$  control for common

macroeconomic events, seasonal trends, and other time-based factors that affect all cities.  $\theta_{ct}$  control for city-specific baseline growth trends (Angrist and Pischke 2008, Goodman-Bacon 2021). For example, local gross domestic product steadily grew and fell from 2012 to 2017 in San Antonio, Texas and Peoria, Illinois, at annual rates of 6.7% and -3.3%, respectively (United States Bureau of Economic Analysis 2019). These commonly-occurring time-varying city-specific differences in baseline growth rates would not otherwise be controlled for by the aforementioned individual and time fixed effects, so omitting them could bias estimates of  $\beta_1$ . As we show after the main results, our results are robust to the functional form used to model these trends – we obtain the same substantive results assuming a quadratic trend in lieu of a linear one, and when we estimate a linear trend using pre-treatment period data then extrapolate this estimate over the post-treatment period. See Web Appendix G for results assuming a quadratic time trend, and Web Appendix H for results using pre-treatment period linear trend extrapolation.

### *City-Level E-Scooter Entry Decision*

One way of assessing the potential for bias arising from time-varying unobservables at the city-level is by analyzing the reasons why cities launch e-scooter programs. To this end, we performed a study of all publicly disclosed announcements in cities launching e-scooter programs (see, for example, City of Atlanta 2019, City of Austin 2018, City of Minneapolis 2019, Lime & City of San Francisco 2018, City of Portland 2019, City of Santa Monica 2019). In these announcements, a variety of purposes were mentioned, including reducing traffic congestion and reducing carbon emissions, neither of which are directly related to restaurant category revenue. This data suggests that cities do not choose whether and when to launch e-scooter programs to increase restaurant sales.

Over 50% of all e-scooter entries in the United States over the 30-month period from the e-scooter industry's inception in September 2017 through February 2020 occurred during summer 2018. The large number of cities choosing to launch e-scooter programs over this time period is a byproduct of regulation, which act as another helpful source of identification. Summer 2018 was the first summer after e-scooter companies began commercial operations,

and as such, e-scooter companies were primarily focused upon growing penetration and revenue. This made e-scooter companies not selective in terms of the cities they were willing to enter. The primary constraint facing the industry was regulatory in nature (i.e., whether cities would allow e-scooter companies to enter). While the first e-scooters were deployed in September 2017, e-scooter companies did so without approval from local governments, and regulatory frameworks governing e-scooter use did not exist. As such, many of these cities subsequently banned e-scooters from operating in their cities (Irfan 2018). By summer 2018, however, regulatory frameworks were successfully put in place, as was an industry-wide data recording standard enabling cities to systematically track e-scooter usage and deployment (e.g., vis a vis maximum e-scooter deployment quotas set by cities) known as the “Mobility Data Specification.” Summer temperatures also made it seasonally attractive to allow e-scooter companies into new cities. These factors drove an upward shock to e-scooter entry in summer 2018, mitigating concerns of anticipatory behavior from cities. Just 15 cities allowed e-scooters to operate over the prior 9 months, and another 29 cities did so over the subsequent 16 months. For additional details, see Web Appendix B.

### *Parallel Pre-Trends*

The canonical parallel trends assumption is that the average outcome among the treated and control groups would have followed parallel trends if the treatment had not occurred. As in prior literature, we first evaluate this through a model-free visualization (Fisher, Gallino, and Xu 2019, Goldfarb, Tucker, and Wang 2022). In Figure 3, we plot average restaurant spending for treatment and control individuals prior to e-scooter entry. It is visually evident that average restaurant spending for both groups is roughly parallel over the pre-treatment period, even before we perform matching, implying that time-varying unobservables at the city- and individual-level are not prominent in the pre-treatment period.

To more formally assess whether time-varying unobservables are present in the pre-treatment period, we perform a regression test to measure differences in spending trends between treated and control groups over the pre-treatment period, as in Bronnenberg, Dubé, and Sanders 2020 and Fisher, Gallino, and Xu 2019. Consider the following regression

Figure 3: Average Restaurant Spend Prior to E-scooter Entry



*Note:* The dotted dark blue line represents the average restaurant spend per panel member (\$) for treated individuals, while the solid light blue line represents the corresponding figure for control individuals. The x-axis range is January 2017 to May 2018, over which period both treated and control cities did not have active e-scooter programs.

equation:

$$(2) \quad \log(\text{Spend}_{it} + 1) = \delta_0 + \delta_1 \text{ScooterAdopted}_{c(i)} + \phi_i + \xi_t + \epsilon_{it}$$

$\text{ScooterAdopted}_{c(i)}$  is a binary variable equal to 1 if the city  $c$  in which individual  $i$  resides allows e-scooters to operate during the observation period and 0 otherwise;  $\delta_1$  represents the deviation from a common time trend for individuals in treatment cities relative to control cities, making  $\delta_1$  the coefficient of interest for this test. We run this regression over a pre-treatment window from January 2017 to May 2018, before e-scooters were introduced in the cities we study. As in Equation 1,  $\text{Spend}_{it}$  represents total restaurant spending for individual  $i$  in week  $t$ . We infer that  $\delta_1$  is not significantly different from 0 at the 10% level ( $p = .19$ ), so we cannot reject the null hypothesis of parallel pretrends.

As mentioned at the beginning of this section, we perform a series of additional analyses to further evaluate whether particular manifestations of time-varying unobservable confounders (e.g., changes in income) may be biasing our results. We discuss these supplementary analyses after presenting our main results in the next section.

## EMPIRICAL RESULTS

In Table 2, we present the results of estimating Equation 1, as well as the corresponding results when our dependent variable is equal to total spending in the fast food and sitdown subcategories. Standard errors in all of the following results are clustered by city, to allow for correlation of errors between individuals within a city. This table shows that the impact of an e-scooter program on overall restaurant spending is both statistically and economically significant, but that this impact differs by category, with a larger effect in the fast food category and a smaller effect in the sitdown category.

Table 2: Estimates of the Effect of E-Scooter Entry on Restaurant Spending

|                   | <i>Dependent variable:</i> |                         |                         |
|-------------------|----------------------------|-------------------------|-------------------------|
|                   | Total                      | Fast Food               | Sitdown                 |
|                   | (1)                        | (2)                     | (3)                     |
| Scooter Entry     | 0.05172***<br>(0.01711)    | 0.04503***<br>(0.01688) | 0.01727***<br>(0.00548) |
| Individual FE     | ✓                          | ✓                       | ✓                       |
| Time FE           | ✓                          | ✓                       | ✓                       |
| Mean of Dep. Var. | 1.726                      | 1.547                   | 0.421                   |
| $R^2$             | 0.338                      | 0.339                   | 0.142                   |
| No. of Obs.       | 7,664,415                  | 7,647,750               | 7,196,145               |

*Note:* Results are obtained from estimating Equation 1, with dependent variables corresponding to total restaurant spending ('Total') and spending within each restaurant subcategory. Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

In Table 3, we show results analogous to those contained in Table 2, except that propensity score matching (PSM, models (1)-(3)) and no matching (models (4)-(6)) are used in lieu of CEM. The estimands using CEM and PSM are similar, but CEM tends to achieve better covariate balance with lower model dependence than PSM. CEM and PSM models include a near-identical number of treated units (13,583 and 13,656 individuals for CEM and PSM respectively), with less than 1% of individuals dropped due to the matching procedure, so the treated populations being analyzed are roughly the same for the two models.<sup>5</sup> The unmatched models do not reduce bias due to confounding through a matched sample, making their estimates less credible. However, the unmatched models do not discard any in-

dividuals, unlike CEM and PSM, broadening their estimands. Given the low rate of dropout due to the matching procedures, this primarily manifested itself through a significantly large number of control individuals being included for the unmatched models.

Table 3: Alternative Matching Procedures

| <i>Matching:</i> | PSM                    |                       |                         | Unmatched               |                         |                         |
|------------------|------------------------|-----------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| <i>DV:</i>       | Total                  | Fast Food             | Sitdown                 | Total                   | Fast Food               | Sitdown                 |
|                  | (1)                    | (2)                   | (3)                     | (4)                     | (5)                     | (6)                     |
| Scooter Entry    | 0.03452**<br>(0.01632) | 0.03018*<br>(0.01594) | 0.01098***<br>(0.00407) | 0.04442***<br>(0.01319) | 0.03808***<br>(0.01274) | 0.01456***<br>(0.00524) |
| Individual FE    | ✓                      | ✓                     | ✓                       | ✓                       | ✓                       | ✓                       |
| Time FE          | ✓                      | ✓                     | ✓                       | ✓                       | ✓                       | ✓                       |
| $R^2$            | 0.373                  | 0.371                 | 0.164                   | 0.349                   | 0.352                   | 0.146                   |
| No. of Obs.      | 4,539,810              | 4,530,900             | 4,244,295               | 15,840,825              | 15,806,835              | 14,967,975              |

*Note:* Results are obtained from estimating Equation 1 using propensity score matching and using no matching, with dependent variables corresponding to total restaurant spending ('Total') and spending within each restaurant subcategory. Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

A comparison of the results in Tables 2 and 3 shows that the estimated effects using CEM, PSM, and unmatched models are comparable to one another. Treatment effect estimates using CEM and PSM methods are within two percentage points of one another in all three categories. All three models infer positive, statistically significant effects in all categories, and smaller effects for the sitdown category relative to the fast food category. The magnitude of the effect sizes are slightly larger (smaller) for CEM (PSM) models than unmatched models, respectively, but in all three models, the magnitudes are economically meaningful for the overall restaurant and fast food categories. This suggests that the substantive conclusions are not sensitive to differences in how aggressively we attempt to reduce bias through different matching methods.

Another concern stems from our using a log transformed dependent variable. We add a constant to each spend before performing the log transformation to ensure that our dependent variable is well-defined in time periods for which spend is equal to 0. We add a constant of 1, but this constant is arbitrary and its value can affect our regression results

by changing the shape of the resulting transformed dependent variable. To address this, we consider a robustness check in which we apply an inverse hyperbolic sine (IHS) transformation to the spend data instead (Aihounton and Henningsen 2021). A favorable property of the IHS transformation is that it is well-defined when spend is equal to or less than 0 without requiring the addition of an arbitrary constant. Table 4 shows that the results using the IHS transformation are consistent with the results using the log transformation from Table 2.

Table 4: Analysis Using Inverse Hyperbolic Sine Transformation

|               | <i>Dependent variable:</i> |                         |                         |
|---------------|----------------------------|-------------------------|-------------------------|
|               | Total                      | Fast Food               | Sitdown                 |
|               | (1)                        | (2)                     | (3)                     |
| Scooter Entry | 0.06203***<br>(0.02046)    | 0.05455***<br>(0.02027) | 0.02058***<br>(0.00653) |
| Individual FE | ✓                          | ✓                       | ✓                       |
| Time FE       | ✓                          | ✓                       | ✓                       |
| $R^2$         | 0.331                      | 0.332                   | 0.141                   |
| No. of Obs.   | 7,664,415                  | 7,647,750               | 7,196,145               |

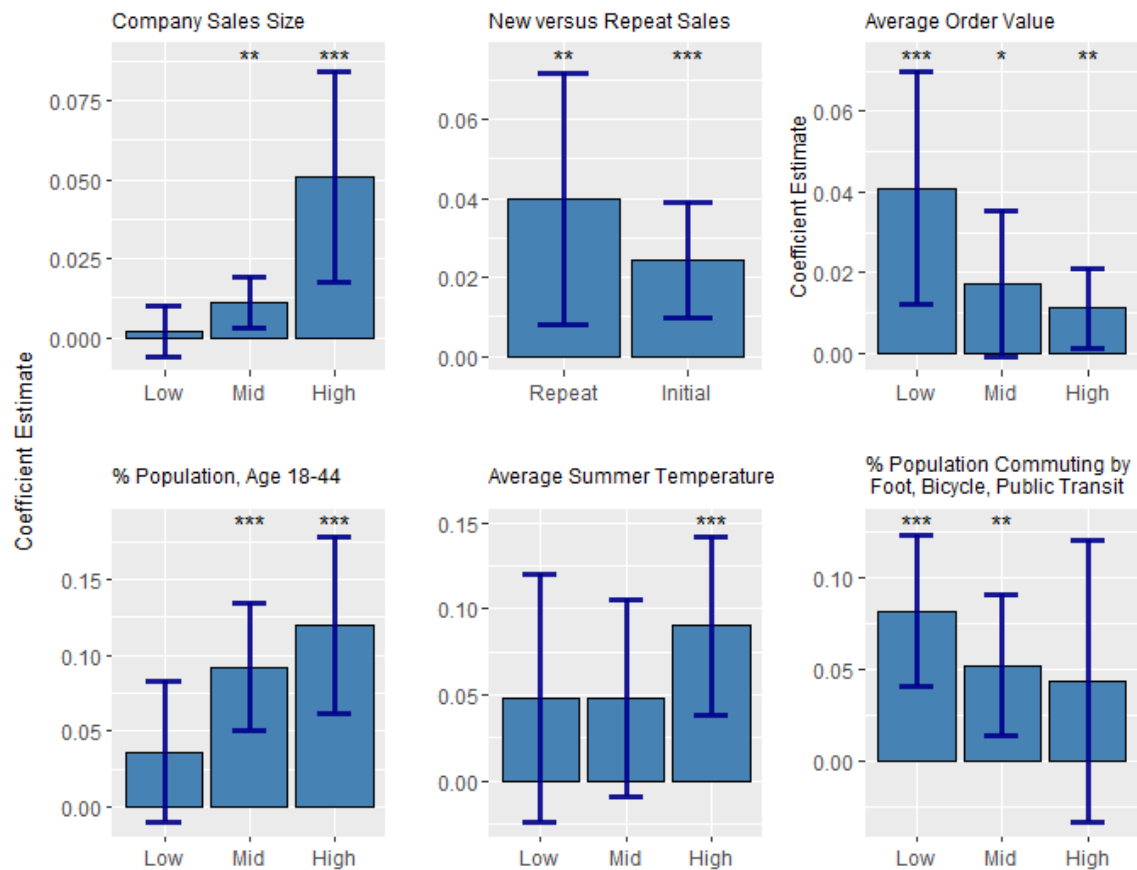
*Note:* Model setting is the same as that of our main analysis, except that we apply an inverse hyperbolic sine transformation to the dependent variable instead of a log transformation. Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

### *Subgroup Analysis for Heterogeneous Treatment Effects*

The impact of e-scooter deployment may be different for companies as a function of company-specific factors, and for cities as a function of city-specific factors. These effects are important to understand, especially for policymakers who may be contemplating the launch of an e-scooter program in their cities, given their cities' characteristics.

We estimate how the causal impact of e-scooter program introduction varies as a function of (1) company sales over the pre-treatment period, (2) company average order value (AOV), (3) sales from new customers versus repeat customers, (4) share of the city's population aged 18 to 44, (5) average summer temperature in the city, and (6) share of the city's population commuting by foot, bicycle, and public transit. Figure 4 visualizes the results, which we describe in more detail next.

Figure 4: Heterogeneous Effect Analysis



*Note:* Each bar represents the causal effect of e-scooter program upon (log transformed) restaurant spending. Dark blue vertical lines represent 95% confidence intervals for each coefficient estimate. 'High', 'Mid', and 'Low' along the x-axis represent the top 0-33%, 33%-66%, 66%-100% percentiles (i.e., tercile splits) by company sales size, average order value, share of age 18-44 population, average summer temperature, and share of commuting by walk, bicycle, and public transit. 'New Versus Repeat Sales' represents coefficient estimates when our dependent variable is equal to total new customer spending (i.e., the sum of all spending made by customers at restaurants they had not previously purchased at before) versus repeat customer spending (i.e., total restaurant spending minus new customer spending). \* p<0.10, \*\* p<0.05, \*\*\* p<0.01

*Company sales size.* The effect of e-scooter entry in a given city may be different for large companies relative to small companies in that city. We assess this by categorizing companies into tercile-based groups based upon sales in each city over the pre-treatment period. We then estimate Equation 1 when our dependent variable is equal to total restaurant spending in each of the resulting subgroups separately. The top-left panel of Figure 4 shows that the impact of e-scooter entry increases monotonically as a function of company size, with significant positive effects for mid-to-large-sized companies.

*New versus repeat sales.* A related factor is whether e-scooter program introduction lifts spending at businesses that consumers have not previously bought from, versus expanding spending from repeat customers. We measure “new sales” by summing spending each week across individuals at all companies that those individuals have not purchased at in any previous week. Conversely, we measure “repeat sales” by summing spending each week across individuals at all companies that have made at least one purchase at in any previous week. The results, depicted in top-middle panel in Figure 4, show that the impact of e-scooter program introduction is positive and statistically significant for both new and repeat spending.

*Average order value.* We assess heterogeneity in the treatment effect as a function of AOV by stratifying companies into tercile-based groups based upon historically observed average spend given purchase in each city. The top-right panel of Figure 4 shows that the impact of e-scooter program monotonically decreases as a function of AOV, with the largest treatment effects accruing to companies with lower AOVs.

*Age.* We next stratify cities into tercile-based groups based on the share of the population whose age is between 18 and 44, as survey reports summarizing e-scooter activity across the U.S. show that more than half (50% to 73%, depending upon the city) of e-scooter riders are under the age of 40 (NACTO 2019). The bottom-left panel of Figure 4 shows that the impact of e-scooter entry is statistically significant in cities with a middle-to-high proportion of young-to-middle aged individuals. In contrast, the effect of e-scooter entry is small and statistically insignificant in cities with a low proportion of young-to-middle aged individuals.

*Summer temperature.* We categorize cities into tercile-based groups based on average summer temperature. The bottom-middle panel of Figure 4 shows no clear directional pat-

tern in estimated effects as we move from low to high temperatures, and the differences across tercile groups are not statistically significant. Differences in absolute statistical significance appear to be driven primarily by variation in statistical power.

*Public transit infrastructure.* We measure public transit infrastructure through the proportion of the population commuting by foot, bicycle, and public transit, as reported by the United States Census Bureau. The bottom-right panel of Figure 4 shows that lift from e-scooter program entry is higher for cities with lower existing public transit infrastructure.

In summary, e-scooter program introduction primarily drives sales lift for larger companies selling at lower price points. It aids both discovery of new restaurants and expansion of business at existing ones. From a city perspective, the launch of an e-scooter program has a larger effect in cities with a medium-to-high proportion of young-to-middle age citizens and less pre-existing public transit infrastructure. It appears to benefit cities of all climates.

#### *All Spending on Local Businesses Except Restaurant Categories*

One concern is that treated individuals may increase their spending at restaurants but decrease it in other segments of the local economy (e.g., due to budget constraints). If this were the case, then the lift figures we estimate may overstate the actual overall increase in spending on the local economy after an e-scooter program is introduced. To evaluate this, we first obtain spending across all in-store purchases, plus spending in the restaurant delivery and grocery delivery categories. We then subtract total in-store restaurant spending (i.e., our focal dependent variable) to derive a measure of total “local” spending excluding in-store restaurant spending. Finally, we re-estimate Equation 1 using local spending excluding in-store restaurant spending as the dependent variable. The results in Table 5 show that the impact of e-scooter program entry on total local spending excluding in-store restaurant spending is positive but not significantly different from 0. This implies that the lift in restaurant sales due to e-scooter program entry is not offset by a compensatory decline in other local spending, even if the latter effect is, in aggregate, not significant. To the extent that budget constraints drive a decrease in spending, then, our results would suggest that this decline is in a “non-local” spending category, such as e-commerce spending.

Table 5: Spending on Local Businesses across All Categories

|               | <i>Dependent variable:</i> |                         |
|---------------|----------------------------|-------------------------|
|               | All but Restaurants        | Restaurants             |
|               | (1)                        | (2)                     |
| Scooter Entry | 0.03818<br>(0.03081)       | 0.05172***<br>(0.01711) |
| Individual FE | ✓                          | ✓                       |
| Time FE       | ✓                          | ✓                       |
| $R^2$         | 0.372                      | 0.338                   |
| No. of Obs.   | 7,695,765                  | 7,664,415               |

*Note:* Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

### *Robustness Checks*

We now turn to additional analyses to further evaluate concerns of time-varying unobserved confounders and assess the robustness of our results in light of these concerns. We first run a series of placebo tests. We then use a permutation test and run variations of our focal model specification to evaluate robustness to other possible forms of endogeneity.

*Placebo tests.* We conduct falsification checks with three placebo categories – grocery delivery (e.g., Instacart), mass transit (e.g., Greyhound Bus), and air travel (e.g., Delta Airlines). We would not expect activity in these categories to be causally related to the introduction of an e-scooter program – for example, it is unlikely that the launch of an e-scooter program will affect individuals’ propensities to engage in out-of-state travel. They can also be helpful for empirically evaluating various forms of time-varying unobservable confounding. For example, if an individual’s income went up in the post-treatment period, and that increase caused e-scooter spending and restaurant spending to go up, this could manifest itself in the form of higher spending in discretionary categories such as grocery delivery. We can evaluate whether these time-varying unobservable confounders may be spuriously driving our results by re-running the regression from Equation 1 with dependent variables corresponding to these placebo categories.

The results in Table 6 show that all placebo categories do not have statistically significant

Table 6: Placebo Tests

|               | Online Grocers       | Mass Transit         | Air Travel           |
|---------------|----------------------|----------------------|----------------------|
|               | (1)                  | (2)                  | (3)                  |
| Scooter Entry | 0.00732<br>(0.02296) | 0.00435<br>(0.01681) | 0.00813<br>(0.00817) |
| Individual FE | ✓                    | ✓                    | ✓                    |
| Time FE       | ✓                    | ✓                    | ✓                    |
| $R^2$         | 0.251                | 0.093                | 0.057                |
| No. of Obs.   | 394,845              | 298,980              | 4,713,060            |

*Note:* Results are obtained by estimating Equation 1, with dependent variables corresponding to the placebo categories shown in the top-most row of the table. Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

relationships with e-scooter entry at the 10% level.

*Permutation tests.* As in Christian and Barrett (2017) and Barron, Kung, and Proserpio (2021), we carry out a permutation test to evaluate how likely our results would have been, had there been no relationship between cities' decisions to introduce e-scooter programs and spending in the restaurant category. Each city is associated with a binary vector representing whether an e-scooter program had already been launched or not at each time point. We randomly permute this vector by city (e.g., randomly assigning Baltimore, Maryland's treatment timing to Arlington, Virginia), while leaving individuals' home city, restaurant spending, and other data unpermuted. We then re-run Equation 1 on the permuted data, obtain the coefficient estimates and corresponding t-statistic values associated with e-scooter program introduction,  $\beta_1$ , and repeat this procedure 200 times.

Permutation by city breaks the association between restaurant spending and e-scooter program introduction, while preserving city-level time trends in spending. Through this procedure, we obtain a distribution of the significance of their association under the null hypothesis that there is no association between the two. If e-scooter program introduction significantly impacts restaurant spending, the t-statistic for  $\beta_1$  in the original unpermuted data would be significantly different from its associated null distribution.

We find that the t-statistic for  $\beta_1$  using the original data is indeed statistically significant vis a vis the null distribution ( $p < .001$ ), implying that our original estimate is significantly

different from what we would have expected if there was no relationship between restaurant spending and e-scooter entry. For further details, see Web Appendix F.

*Variation in model specification.* To assess whether our results are an artifact of the exact specification of our model, or may be at the risk of bias due to specific forms of endogeneity, we consider the following alternative specifications:

1. Different time trends: a potential concern is that our estimate is biased due to our assumption that city-specific time trends are linear if they are in fact non-linear. To mitigate this concern, we run a robustness check assuming a quadratic time trend (Table W 5 in Web Appendix G).
2. Alternative time trend specification: if treatment effects or unobservables were time-varying, estimating time trend variables upon all of the data (i.e., both the pre-treatment and post-treatment periods) could bias our estimates (Goodman-Bacon 2021). To address this, we run an alternative model specification in which the coefficients associated with the time trend are estimated using the pre-treatment period data alone, and then extrapolated into the post-treatment period, as this procedure is not biased by treatment effects or unobservables that vary over time (Web Appendix H).
3. Different unit of time: treatment effects may vary depending on the unit of time that is used due to time aggregation (M. Kim, Bradlow, and Iyengar 2022). To evaluate concerns that our results are an artifact of our having chosen a week unit of time, we consider models in which a daily unit of time or monthly unit of time are used instead (i.e., modeling purchase-days or purchase-months instead of purchase-weeks). Our estimate could also be biased because cities introduced e-scooter programs at different times (i.e., while all treatments occur within summer 2018, the exact timing of these treatments varies by city). To address this, we re-run our main model specification with a five-month unit of time, such that the treatment window is collapsed into a single time period. As a result, there is a pre-treatment period, one time point over which the e-scooter program introduction event occurs (conservatively excluded from the post-treatment period), followed by a post-treatment period (Table W 5 in Web

Appendix G).

The results of these and other checks are substantively the same as those of our main analysis in terms of magnitude and direction. In every alternative scenario, inclusive of those mentioned after the main results, the estimated treatment effect for overall restaurant spending is positive and significant at the 5% level. Weekly lift estimates range from 3.5% to 8.0%, relatively close to the 5.2% lift estimate we obtain using our main model specification. The consistency of the results under these alternative specifications to those of our main model suggests that our results are robust to the identification threats summarized above. While we cannot rule out the possibility that other time-varying confounders may bias our results, the analyses we perform above suggest that the main results appear robust to a variety of different ways in which those concerns could have manifested themselves.

### *POLICY IMPLICATIONS*

To further illustrate the economic impact of the results, we provide a conservative estimate of the population-level dollar amount of spending lift created by the introduction of these e-scooter programs.

We can estimate incremental restaurant sales from treated individuals for our 49 treated cities over the universe of restaurant companies tagged by the data provider in the post-treatment period after e-scooter program introduction, all else equal, through the coefficient estimates from our main model in Equation 1 (summarized in Table 2). For each treated individual over each post-treatment time period, we use Equation 1 to estimate total sales with versus without e-scooter entry by setting the treatment variable equal to 1 versus 0, respectively, while holding all parameters equal to their coefficient estimates. We then sum expected spending with versus without e-scooter program introduction across treated individuals over a one-year post-treatment period, then take the difference.<sup>6</sup> This represents the estimated dollar uplift in spending from treated individuals (and thus overall, assuming no treatment effect for untreated individuals) within the credit card panel, across the 391 companies covered by our data provider in the restaurant category, in the 49 treated cities we study. The credit card panel represents approximately 4.1% of the working age and older

population across all treatment cities, so we gross up this figure by a factor of 24.58 to obtain our estimate of population-level dollar lift for the 391 restaurant companies. This implies that all else equal, approximately \$11.3 million in incremental spending over a one-year period was created for these restaurant companies in our 49 treated cities because of the introduction of e-scooter programs.

We also estimate how much spending was lifted per e-scooter allowed to operate by the cities. We first identified every city for which citywide e-scooter allowed data was available, either through a government website or a local news source. Of the 49 cities we study, we were able to obtain citywide e-scooter allowed data for 32 of them. Table W 6 in Web Appendix I lists each city and the corresponding number of e-scooters allowed for that city, when available. 58,380 e-scooters in 32 cities were permitted to operate after e-scooter entry across these treatment cities. Taking the ratio of total population-level spending lift to e-scooters allowed across all cities we have permitted e-scooter fleet size data for, we estimate that \$179.1 in restaurant sales was created per e-scooter allowed to operate.

As noted in the introduction, these calculations assume that the panel we analyze is representative of the population – that the panel represents approximately 4.1% of the population, and that treated individuals observed in the panel are representative of treated individuals not observed in the panel. We have shown that restaurant spending trends in the panel correlate strongly with restaurant spending trends at the population-level, as observed through public company disclosures (Web Appendix D), giving us some empirical evidence supporting these assumptions. However, we acknowledge that the actual population-level impact could be higher or lower to the extent that these assumptions are violated.

Finally, we can compare how the benefits of these programs – to the city directly and to the restaurant sector more broadly – compare to the cost to the city of rolling out the programs. Currently, cities pass on the cost of roll-out onto e-scooter companies through fees (e.g., for permitting and monitoring), making total fees a reasonable initial estimate for the roll-out cost to the city. While fee data is sparse, the city of Dallas reported that it had collected \$67,848 in e-scooter-related fees in 2019, so we carry out these calculations for Dallas<sup>7</sup>. Of the aforementioned total lift across all cities of \$11.3 million, \$423 thousand of

this accrued to Dallas. Consider benefits to the city directly, versus to the restaurant sector:

1. The local sales tax rate in Dallas is 8.25%, implying that the city generated at least \$34,896 in incremental sales tax revenue due to e-scooter rollout, or 51.4% of the cost of rollout to the city.
2. We may consider the benefit to the restaurant industry in terms of the incremental revenue and profitability that is created through e-scooter rollout. As mentioned above, e-scooter rollout created \$423 thousand in incremental revenue for the restaurant sector annually. When we multiply this dollar lift estimate by the contribution margin in the restaurant sector, as estimated through filings with the Securities and Exchange Commission by major publicly-traded companies in these categories, we estimate \$94 thousand in incremental profits to the restaurant sector annually, or approximately 139% of the cost of rollout to the city. For details, see Web Appendix J.

In sum, we infer that the lift created by e-scooter deployments to the city, and to the restaurant sector, are meaningful relative to the cost of rolling out the e-scooter program, which is currently absorbed by the e-scooter companies themselves. These figures may prove to be conservative, as the cost of an e-scooter program may be higher in the first year a city rolls out e-scooters than in subsequent years, depressing first-year profitability. Additionally, as mentioned earlier, we only measure spending lift at restaurant companies tagged by the data provider, implying that lift across all firms in the restaurant sector (i.e., tagged by the data provider or not) are likely to be larger than the figures provided here. As such, these figures may be a lower bound on the economic benefits that may accrue to policymakers in cities such as Dallas.

## *DISCUSSION*

Our study suggests that micromobility creates positive economic externalities for the local economy they operate in – introducing an e-scooter program lifts sales in the restaurant sector, and this lift is not offset by a corresponding decline in non-restaurant local sales. The findings represent an opportunity for restaurant operators and e-scooter companies to

capitalize on the economic surplus that they create. For example, these spillovers could be further facilitated through location-based advertising opportunities (Molitor et al. 2020), which could serve as a valuable additional revenue source for e-scooter companies. Likewise, these results suggest that city governments looking to stimulate the local economy may be incentivized to take a more lenient stance towards e-scooter usage.

We readily acknowledge that this research comes with limitations. Given that the level of geographical aggregation of our main data source is at the city-level, we can provide the average overall impact of introducing an e-scooter program in a city, but are not able to measure effects at a finer geographical unit. While a finer-grained analysis may be harder to act upon as e-scooter introduction is inherently a city-level decision, such an analysis may still be of interest to policymakers. We also acknowledge that our transactional data, while rich, provides us limited visibility into the mechanisms driving spending lift. Incorporating first-party data sources that could allow for a deeper understanding of mechanisms is an interesting area for future work. We also reiterate that time-varying unobservable sources of individual-level and city-level confounding may be present. While the various analyses we conducted support the notion that many specific threats to identification do not appear to be first-order concerns, we cannot rule out that there may be other threats that are.

There are several other questions regarding the effect of e-scooter-enabled mobility on company sales and consumer behavior that we also leave for future work. For example, while our analysis only covers non-restaurant companies through an aggregated “total local spending excluding in-store restaurant spending” category, decomposing this further into subcategories such as entertainment (e.g., movie theaters) and health/fitness (e.g., gyms), could provide deeper insight into heterogeneity in the effects that e-scooter programs have across a wider range of categories. It could also be valuable to understand if our results generalize to geographies outside of the United States. While we believe our results should extend to other geographies, empirical proof that they do could provide further confidence to international policymakers. Finally, while we focus on spending from individuals in their inferred home locations, it could be valuable to analyze how e-scooters change the behavior of tourists.

In conclusion, we believe that micromobility is an understudied area within marketing, and that studying its economic impact is important for a variety of stakeholders, including city policymakers, consumers, companies, and e-scooter companies themselves. We hope that this work represents a first step towards better understanding these important topics.

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### *Notes*

<sup>1</sup><https://nacto.org/2020/08/27/136-million-trips-taken-on-shared-bikes-and-scooters-across-the-u-s-in-2019/>;  
<https://nabsa.net/about/industry/>

<sup>2</sup><https://mobilityforesights.com/product/scooter-sharing-market-report/>

<sup>3</sup>These figures may understate the total number of trips that entailed going to or from a restaurant if going to or from a restaurant was a secondary purpose.

<sup>4</sup>There were a total of 54 cities that launched e-scooter programs from June to October 2018, but five were dropped due to our matching procedure.

<sup>5</sup>Differences in sample size are due to CEM matching more control individuals to treated units than PSM.

<sup>6</sup>We restrict our analysis to retrospective and not predictive estimates because we include time fixed effects in our model.

<sup>7</sup><https://dallascityhall.com/departments/transportation/traffic-calming/DCH%20Documents/Dockless%20Vehicle%20Permit%20Application%20July%202018.pdf>

# Wheels to Meals: Measuring the Impact of Micromobility on Restaurant Demand

## Online Appendix

May 2023

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### *Web Appendix A: Companies Included in Analysis*

We obtain transaction data for the companies listed below. Fast Food and Sitdown are subcategories within the focal restaurant category. Online Grocers, Air Travel, and Mass Transit are categories used to perform placebo tests in the Model Development section of the main paper. Other subcategories are used for constructing matching variables. Companies are listed in alphabetical order.

#### *Companies in Fast Food Category*

&pizza, 1000 Degrees Pizza, A&W Restaurants, Ample Hills Creamery, Arby's, Au Bon Pain, Auntie Anne's, B.GOOD, Baja Fresh, Ben & Jerry's Ice Cream, Bertucci's, Blaze Pizza, Blue Bottle, Bluestone Lane, Bojangles', Boloco, Boston Market, Braums, Bruegger's, Bruxie, Burger King, by CHLOE, Cafe Rio, Captain D's, Caribou Coffee, Carl's Jr., Carvel, Cava Grill, Charleys Philly Steaks, Checkers, Chicken Express, Chick-fil-A, Chipotle, Chopt Salad, Chuck E. Cheese's, Church's Chicken, Cicis, Cinnabon, Cold Stone Creamery, Cook Out, Coolhaus, Corner Bakery, Cusi, Costa Vida Fresh Mexican Company, Culver's, Dairy Queen, D'Angelo Grilled Sandwiches, Del Taco, DiBella's, Dig Inn, Domino's Pizza, Donatos Pizza, Dos Toros, Dunkin' Brands, Dutch Bros. Coffee, Eggslut, El Pollo Loco, Fazoli's, FIKA, Financier Patisserie, Firehouse Subs, Five Guys, Foosackly's, Fox's Pizza Den, Freddy's Frozen Custard & Steakburgers, Freebirds World Burrito, fresh & co, Fresh Brothers, Friendly's, Fuzzy's Taco Shop, Georgetown Cupcake, Godfather's Pizza, Golden Chick, Golden Corral, Great American Cookie, Gregorys Coffee, Habit Restaurants, Hale and Hearty, Hardee's, honeygrow, Hopdoddy Burger Bar, Hot Dog on a Stick, Hu Kitchen, Hungry Howies, Ike's Place, Imo's Pizza, In-N-Out Burger, Insomnia Cookies, It's Just Pasta, Jack in the Box, Jamba Juice, Jason's Deli, Jeni's Splendid Ice Creams, Jersey Mike's Subs, Jet's Pizza, Jimmy Johns, Joe & The Juice, Juice Generation, Juice Press, Just Salad, KFC, Kreation Organic Juicery, Krispy Kreme, Kura Sushi USA, La Colombe, La Madeleine, Le Pain Quotidien, Ledo Pizza, Lemonade LA, Lenwich, Levain Bakery, Little Caesars, Luna Grill, MAD Greens, Maggie Moo, Maison Kayser, Maple Street Biscuit Company, Marble Slab Creamery, Marco's Pizza, Mazzio's Italian Eatery, McAlister's Deli, McDonald's, Mendocino Farms, Mezeh, MISSION BBQ, Mixt, MOD Pizza, Modern Market, Moe's Southwest Grill, Monical's Pizza, Mountain Mike's Pizza, Naf Naf Grill, Naya, Newk's Eatery, Noodles, Pancheros Mexican Grill, Panda Express, Panera Bread, Papa Gino's, Papa John's, Papa Murphy's, Paradise Bakery & Cafe, Partner's Coffee, PDQ, Peet's Coffee, Penn Station, Peter Piper Pizza, Philz Coffee, Piada Italian Street Food, Pie Five, Pieology, Pinkberry, Pita Pit, Pizza Hut, Pizza Inn, Pizza Patron, PizzaRev, Pizzeria Locale, Pluckers Wing Bar, Pokeworks, Pollo Tropical, Popeyes Louisiana Kitchen, Potbelly, Pret A Manger, Pretzelmaker, Protein Bar, Qdoba, Quiznos, Raising Cane's, Rally's, Roberto's Tacos, Rosa's Cafe, Rosati's Pizza, Roti Modern Mediterranean, Round Table Pizza, Ruby Slipper, Rusty Taco, Sbarro, Schlotzsky's, Seattle's Best, Shake Shack, ShopHouse, Sightglass Coffee, Slim Chickens, Smashburger, Smitten Ice Cream, Smoothie King, Snap Kitchen, Sonic Drive-In, St. Louis Bread Company, Starbucks, Steak 'n Shake, Subway, SusieCakes, sweetgreen, Taco Bell, Taco Bueno, Taco Cabana, Tacodeli, Tasty Made, Teavana, Tender Greens, The Coffee Bean & Tea Leaf, Thighstop, Think Coffee, Tim Hortons, Toppers, Torchys, Tropical Smoothie, Uncle Maddio's, Veggie Grill, Wendy's, Wetzel's Pretzels, Whataburger, Which Wich, White Castle, Wichcraft, Wingstop, Xi'an Famous Foods, Your Pie, Zaxby's, Zoe's Kitchen

### *Companies in Sitdown Category*

5 Napkin, 99 Restaurant & Pub, Another Broken Egg, Anthony's Coal Fired Pizza, Applebee's, Bahama Breeze, Bar Louie, Barcelona Wine Bar, Bartaco, Beef O'Brady's, Belcampo Meat Co, Benihana, BJ's Restaurants, Black Angus Steakhouse, Black Bear Diner, Bob Evans, Bombshells, Bonefish Grill, Border Cafe, Breakfast Republic, Brio Tuscan Grille, Bubba Gump Shrimp, Bubba's 33, Buca di Beppo, Buffalo Wild Wings, BurgerFi, Burtons Grill, Cafe Gratitude, California Fish Grill, California Pizza Kitchen, Capital Grille, Carrabba's Italian Grill, Chart House, Cheddar's Scratch Kitchen, Cheeseburger in Paradise, Cheesecake Factory, Chicken 'n Biscuits, Chicken Sammy's, Chili's, Chuy's, Claim Jumper, Columbus Fish Market, Cooper's Hawk Restaurant, Cooper's Hawk Wine Club, Cotton Patch Cafe, Cracker Barrel, Del Frisco's, Denny's, Eddie V's, El Torito, Famous Dave's, Famous Toastery, Firebirds Wood Fired Grill, First Watch, Fleming's Prime Steakhouse & Wine Bar, Flower Child, Flying Biscuit, Fogo de Chao, Fox Restaurant Concepts, Fresh Set, Fudruckers, Genghis Grill, Gloria's Latin Cuisine, Gordon Biersch, Grand Lux, Great American Buffet, Hard Rock Cafe, Hillstone, HipCityVeg, Hooters, Houlihan's, IHOP, Islands Finge Burgers & Drinks, It's Just Wings, J. Alexander's, Joe's American, Joe's Crab Shack, Johnny Carino's, Johnny Rockets, Keke's Breakfast Cafe, Kitchen 1883, Kona Grill, Landry's Seafood House, Legal Sea Foods, Logan's Roadhouse, LongHorn Steakhouse, Luby's Cafeteria, Lucille's Smokehouse BBQ, Maggiano's Italian Classics, Maggiano's Little Italy, Marie Callender's, Mastro's, Max & Erma's, Max Brenner, McCormick & Schmick's, Mellow Mushroom, Metro Diner, Mi Cocina, Miller's Ale House, Mimi's Cafe, Mitchell's Fish Market, Momofuku, Morton's The Steakhouse, MrBeast Burger, North Italia, Ocean Prime, O'Charley's, Old Chicago Pizza Taproom, Olive Garden, On The Border Mexican Grill & Cantina, Original Pancake House, Outback Steakhouse, P.F. Chang's, Pancake Kitchen, Pappadeaux Seafood Kitchen, Pappasito's Cantina, Pasqually's Pizza, Pei Wei, Portillo's, Quaker Steak & Lube, Quincy's Family Steakhouse, Rainforest Cafe, Red Lobster, Red Robin, Romano's Macaroni Grill, Rosa Mexicano, Ruby Tuesday, Russo's, Ruth's Chris, Salsarita's Fresh Mexican Grill, Saltgrass Steak House, Seasons 52, Shari's Cafe and Pies, Shula's Steak House, Silver Diner, Sizzler, Slo Roast, Smith & Wollensky, Smokey Bones Bar & Fire Grill, Snooze Eatery, SUGARFISH, Sullivan's, Tender Shack, Texas de Brazil, Texas Roadhouse, TGI Fridays, The Burger Den, The Egg & I, The Meltdown, The Melting Pot, The Palm Restaurant, The Wing Dept., Tijuana Flats, Tilted Kilt Pub & Eatery, Tocaya, True Food Kitchen, Twin Peaks, Uncle Julio's, UNO Pizzeria & Grill, Urban Plates, Velvet Taco, Village Inn, Waffle House, wagamama, Western Sizzlin', Wolfgang's Steakhouse, Wood Grill Buffet, World of Beer, Yard House

### *Companies in Online Grocers Category*

1520min, Amazon Fresh, Boxed, Brandless, ButcherBox, Buyk, Cornershop, Daily Harvest, Farm Fresh To You, Farmstead, Favor, FoodKick, Fresh Direct, Fridge No More, Good Eggs, goPuff, Gorillas, GrubMarket, HEB To You, Hungry Harvest, Hungryroot, Imperfect Foods, Instacart, JOKR, Kroger Ship, Mercato, Misfits Market, Peapod, Schwan's, Shipt, The Good Kitchen, Thrive Market, Weee!, Wild Fork Foods

### *Companies in Mass Transit Category*

Amtrak, Bay Area Rapid Transit, BoltBus, Bridges and Tunnels, Chicago Transit Authority, FlixBus, Greyhound Lines, LIRR, Massachusetts Bay Transportation Authority, Megabus, METRO, Metro

North, Metrocard, Virgin Trains, Washington Metropolitan Transit Authority, Washington State Ferries

### *Companies in Air Travel Category*

AirTran, Alaska Airlines, Allegiant Air, American Airlines, Copa Airlines, Delta Airlines, Frontier Airlines, Hopper, JetBlue, Southwest, Spirit Airlines, TSA PreCheck, United Airlines

### *Companies in Restaurant Delivery Category*

Amazon Restaurants, Bite Squad, Caviar, Chowbus, DashMart, Delivery.com, Delivery Dudes, DoorDash, Eat24, EatStreet, ezCater, Fantuan, Food Dudes Delivery, Fooda, Foodler, Foodsby, Goldbelly, GrubHub and Seamless, HungryPanda, MealPal, Munchery, Olo, OrderUp, Postmates, Ritual, SkipTheDishes, Slice, Tapingo, Thistle, UberEATS, Waitr

### *Companies in Ride Share Category*

Juno, Lyft, Uber, Uber Rides, Via

### *Companies in Gas Station Category*

7-Eleven (Fuel), BJ's Fuel, Booster Fuels, BP, Buc-ee's, Bucky's Convenience Stores, Casey's General Store, ChargePoint, Chevron, Circle K, Conoco, Costco Gas, Cumberland Farms, Exxon Mobil, Fas mart, Filld, Flash Foods, GetGo, Gulf Oil, Holiday Station Stores, Kangaroo Express, Kum & Go, Kwik Shop, Kwik Shop-EG Group, Kwik Trip, Loaf N' Jug, Loaf N' Jug-EG Group, Love's, Lukoil, MAPCO, Maverik, Murphy USA, Phillips 66, Pilot Flying J, Quik Stop Markets, Quik Stop-EG Group, QuikTrip, RaceTrac, Raley's Gas, Royal Farms, Sheetz, Shell, Speedway, Sunoco, Texaco, Valero, Wawa

### *Companies in Events & Attractions Category*

AMC, Acme Technologies, Adventure Island, Alamo Drafthouse Cinema, AMF, Animal Kingdom, Aquatica, Atom Tickets, AXS, Beau Rivage, Bellagio, Bowlero, Bowlmor, BrunswickZone, Busch Gardens, Carmike, Carowinds, Castaway Bay, Cedar Point, Century Theatre, Cinema World, Cinemark, Circus Circus Las Vegas, Darien Lake, Dave Buster's, Discovery Cove, Disney, Disney's Hollywood Studios, Dorney Park, Drive Shack, Epcot, Eventbrite, Excalibur, Fandango, Frontier City, Gametime, Gilroy Gardens, Gold Strike, Golfnow, Great America, Great Wolf Lodge, Groupon, Harlem Globetrotters, Hunt A Killer, Ikon Pass, Kalahari Resorts, Kings Dominion, Kings Island, Knotts Berry, Legoland, Legoland Discovery, Liftopia, Live Nation, LivingSocial, Lucky Strike, Luxor, Madame Tussauds, Magic Kingdom, Main Event, Mandalay Bay, Marcus Theaters, MGM Grand, MGM National Harbor, MGM Springfield, Michigans Adventure, Mountain Collective, MoviePass, Peak Resorts, Pink Jeep Tours, Pinstripes, Punch Bowl Social, Rave, Reading International, Regal Cinemas, Round One, San Diego Zoo, Sands Expo Center, Schlitterbahn, Sea Life,

SeatGeek, Seaworld, Sesame Place, Showcase Cinemas, Sinemia, Six Flags, Six Flags Hurricane Harbor Concord, StubHub, The Mirage, The Venetian Resort, Ticketfly, Ticketmaster, Tickets.com, TickPick, Tinseltown USA, tixr, TodayTix, Topgolf, Universal Studios, Vail Lift, Vail Other, Vail Rental, Vail Ski School, Valleyfair, Viagogo, Vivid Seats, Water Country, Wet 'n Wild Orlando, Wet 'n Wild Phoenix, Wet 'n Wild Splashtown, White Water Bay, Worlds of Fun, Wynn Resorts

### *Companies in Home Entertainment Category*

360training, ABCmouse, AbeBooks, Aceable, Affinity Apps, Amazon Music, Aqua-Tots, Audible, Audiobooks, Babbel.com, Badoo, Bandai Namco Entertainment, Bandcamp, Barnes Noble, Barstool Sportsbook, Bartleby, Basic, BBPeopleMeet, Betfair, BetMGM, Big Fish Games, BlackPeopleMeet, Blizzard Entertainment, Blurb, Books-A-Million, Bright Horizons, Bumble, CareerStep, CelerX, Cengage, Chatbooks, Chegg Other, Chegg Services, Children's Courtyard, Christianbook.com, ChristianMingle, Class101, College Board, Community, Course Hero, Coursera, Creative Kids Learning, CrowdStar, Crunchyroll, CuriosityStream, DataCamp, Deezer, Delightful, Diamond Mind, DianDian Interactive, Discord, Discovery Plus, Disney+, Domestika, Dote Shopping, DraftKings, DragonBox, Duolingo, EA Access, EA Mobile Games, eharmony, Electronic Arts, EliteSingles, Epic Games Other, Epic Games Store, ESPN+, Everbrook Academy, Facebook Other, Faithlife, FanDuel, Financial Times, FloSports, Fortnite, FOX Bet, fuboTV, Gaia, Galvanize, Game Show Network, GameFly, Games Workshop, GameStop, Garena, Glu, GooglePlay Music, Gram Games, Grand Canyon University, Grindr, Half Price Books, HBO, Higher Education Services, Hinge, HowAboutWe, Hulu, Huuuge Global, iHeartMedia, IMDb, Inc., iTunes, Jam City, JDate, K12 Inc., Kabam Games, KinderCare, Kindle, King, KingsGroup, KiwiCo, Kobo, La Petite Academy, Lakeshore Learning, LatinoPeopleMeet, LDSPlanet, League of Legends, Lilith Games, LinkedIn, Living Scriptures, Los Angeles Times, Lovevery, Lynda, Magic: The Gathering Arena, MasterClass, Match.com, Mathway, McGraw-Hill Education Inc., Medium, miHoYo, Mindvalley, Montessori Unlimited, Monthly, Moon Active, NaturalMotion, Netflix, Netmarble, Next Big Idea Club, NFHS Network, Niantic, Nintendo, NY Times, Oculus, OkCupid, OnlyFans, Open English, Origin.com, OurTime, Outschool, Pandora Internet Radio, Paramount+, Peacock, Peak Games, Penn Foster, People Magazine, PetPeopleMeet, Phantom EFX, Philo, Pinterest, Plarium, Playdemic, Playstation Network, Playstudios, Playtika, PlentyOfFish, Plex, Pluralsight, Precision Nutrition, Pure Flix, Quizlet, Razer, Readly, Redbox, Riot Games, Roblox, Rockstar Games, Rollic Games, Rosetta Stone, Rovio, Scholastic, Scopely, Showtime Networks, Singlesnet, SiriusXM, Skillshare, Skillz, Slacker Radio, Slader, Sling TV, Small Giant Games, Smule, Snapchat Advertisements, Socialpoint, SoundCloud, Southern Learning, Speeddate.com, Splice, Spooky Cool Labs, Spotify, Stadia, Steam, Study.com, Supercell, Sylvan Learning, TADS, TCGplayer, Teachable, Teachers Pay Teachers, Texture, The Athletic, The Economist, The Keystone School, The Wall Street Journal, The Washington Post, Thinkific, Tidal, Time Magazine, Tinder, TiVo, Tonebase, Tutor Time, TVG Network, Twitch, Twitch Prime, Twitter Advertisements, Tynker, Ubisoft, Udacity, Udemy, University of Phoenix, Univision Now, Unreal Engine, Upgraded, Uplay Plus, USA TODAY, Varsity Tutors, VidAngel, Vimeo, Vudu, World Wrestling Entertainment, WorldWinner, WynnBET, Wyzant, Xbox Game Pass Ultimate, Xbox Live Gold, Xbox Store, Yousician, YouTube, Zoosk, Zwift, Zynga

*Web Appendix B: Treated and Control Cities*

Table W 1: List of Treated and Control Cities

| Scooter Adoption | Treated cities                                                                                                                                                                                                              | Control cities                                                                                                                                                                                                               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June 2018        | Arlington, VA; Baltimore, MD;<br>Dallas, TX; Indianapolis, IN;<br>Milwaukee, WI; Salt Lake City, UT;<br>Santa Barbara, CA; South Lake<br>Tahoe, CA                                                                          | Bethlehem, PA; Germantown, MD;<br>Jacksonville, NC; Madison, WI;<br>Springfield, OH; Sunrise, FL; Wichita<br>Falls, TX; Wilson, NC                                                                                           |
| July 2018        | Cincinnati, OH; Columbus, OH;<br>Detroit, MI; Kansas City, MO;<br>Louisville, KY; Minneapolis, MN;<br>Portland, OR; Raleigh, NC; St. Louis,<br>MO; St. Paul, MN                                                             | Akron, OH; Arlington, TX; Bowie,<br>MD; Chesapeake, VA; Concord, NC;<br>Newport News, VA; Rockville, MD;<br>Roseville, MI; Sarasota, FL;<br>Thousand Oaks, CA                                                                |
| August 2018      | Abilene, TX; Athens, GA; Cleveland,<br>OH; Columbia, MO; Coral Gables,<br>FL; Fresno, CA; Greensboro, NC; La<br>Mesa, CA; Long Beach, CA;<br>Lubbock, TX; Norman, OK;<br>Oklahoma City, OK; Richmond, VA;<br>Stillwater, OK | Champaign, IL; Charleston, SC;<br>Columbia, MD; Columbus, GA; El<br>Centro, CA; Erie, PA; Fayetteville,<br>NC; Fort Wayne, IN; Gastonia, NC;<br>Melbourne, FL; Mobile, AL;<br>Overland Park, KS; Rapid City, SD;<br>York, PA |
| September 2018   | Ann Arbor, MI; Bloomington, IN;<br>Durham, NC; East Lansing, MI;<br>Russellville, AR; Santa Clara, CA;<br>Santa Cruz, CA; Statesboro, GA;<br>Tacoma, WA                                                                     | Bowling Green, KY; Buckeye, AZ;<br>Canton, OH; Des Moines, IA;<br>Greenville, NC; Greenwood, IN;<br>Joliet, IL; Minot, ND; Pittsburgh, PA                                                                                    |
| October 2018     | Boise City, ID; Cary, NC; Lansing,<br>MI; Oxford, OH; Peoria, IL;<br>Providence, RI; Riverside, CA; San<br>Antonio, TX                                                                                                      | Allentown, PA; Augusta, GA;<br>Brandon, FL; Frederick, MD;<br>Gaithersburg, MD; Lancaster, PA;<br>Montgomery, AL; Palm Coast, FL                                                                                             |

There is no exhaustive data set containing all cities that introduced e-scooter programs along with the dates that those programs were first introduced. We create this data set by first merging data from three sources: an internal data set from the North American Bikeshare & Scootershare Association (NABSA), NBC News,<sup>1</sup> and an internal data set from Bird. These data sources often omitted cities contained by other data sources, and reported e-scooter entry dates that conflicted with other data sources. To correct for these issues, we

<sup>1</sup><https://www.nbcnews.com/news/us-news/e-scooters-expansion-map-n998106>

conducted internet searches of local news articles reporting on e-scooter activity for every city in the merged data set. Data is collected at a monthly level of temporal granularity.

We limit our focus to cities which first rolled out e-scooters involving at least one of the four major e-scooter operators – Bird, Lime, Uber, and Lyft – between June 2018 and October 2018. There were 54 cities that first introduced e-scooter programs over summer 2018. 5 cities were then dropped due to our matching procedure, leaving us with the 49 cities listed in Table W 1.

### *Web Appendix C: Process of Filtering Data*

Our data consists of individuals from Earnest Analytics’s “consistent shopper panel.” Consistent shoppers are defined using proprietary logic from Earnest Analytics to identify panel members who are inferred to use credit and debit cards on record for most of their purchases. To identify treatment and control units, we first infer the modal home location (at the city-level) for each member of the consistent shopper panel. We do so by identifying all offline transactions with directly observable (or “known”) location information (i.e., transactions associated with categories such as restaurants, grocery stores, gyms, gas stations, and pharmacies for which the city and state associated with the transaction are directly observable through the data associated with the credit card transaction). Using these known location transactions, we calculate the frequency with which each individual is in each city over the observation period. The modal city (i.e., the city with the highest frequency) for each individual is designated to be that individual’s modal home location. We drop individuals for whom there is more than one city that shows up with the highest frequency (i.e., the highest frequency is exactly the same for more than one city), as we would not be confident in the home location for these individuals. Our results are not sensitive to this assumption, with 1.2% of individuals excluded due to this filter. We then filter out all purchase activity occurring on days in which an individual’s modal location that day is not the same as her modal home location to remove possible distortion that may result from purchase activity while individuals are not in their modal home location (e.g., are on vacation).

We further filter down to individuals whose modal home location falls within one of the 98 treatment and control cities listed in Web Appendix B. These individuals are then matched using the procedure described in Web Appendix E.

### *Web Appendix D: Representativeness of Transactional Data*

In this section, we provide further detail regarding the representativeness analysis discussed in the Data section of the main paper. Many publicly traded restaurant companies regularly disclose revenue, store sales, and/or same store sales data through press releases and filings with the Securities and Exchange Commission (SEC).

In some instances, more than one sales measure is available for a particular company, and in others, more than one brand may be owned by a particular parent company. We select evaluation metrics and brands through the following logic:

- Sales data is available for all public companies, but many public restaurant companies own portfolios of restaurant brands. For example, Restaurant Brands Inc. (ticker: QSR) owns Burger King and Popeye's. Overall company revenue data is readily available for Restaurant Brands, but the company also discloses same-store sales growth figures for Burger King and Popeye's separately. In cases such as this one, we use same store sales data at the brand level as our evaluation metric, instead of overall sales data at the parent company level, to carry out a more granular validation.
- Store sales data is available for many companies, in addition to overall sales data. The former measure represents sales at stores (either company-owned, or company-owned plus franchise stores), while the latter measure includes other revenue sources including the sale of raw ingredients from the parent company to franchise stores. As such, store sales is a more direct measure of end customer spending than overall sales, so when both figures are available, we use store sales and not overall sales as our evaluation metric.
- Some companies report both same store sales growth (also known as comparable store sales growth) and store sales. The former measure is equal to the percentage growth in revenue from stores that have been open for at least one year, relative to sales from those same stores over the same period of time in the previous year. As such, same store sales growth is a measure of "organic" revenue growth, unaffected by revenue from new store openings. In contrast, store sales is equal to total revenue at stores,

regardless of how long those stores have been open for. Same store sales growth is a less comparable measure to our data than store sales, because we cannot identify individual stores (and thus, whether those stores have been open for more than one year). For this reason, when same store sales and store sales are both available, we use store sales and not same store sales as our evaluation metric.

- Sales, store sales, and same store sales are evaluation metrics for 13, 8, and 15 restaurant brands, respectively. For comparability across measures, we operationalize sales and store sales measures as year-on-year percentage change figures, because same store sales (the modal evaluation metric) is only reported as a year-on-year percentage change figure.

After selecting an evaluation metric for each restaurant brand, we collect all available company-disclosed population-level data for it from the first calendar quarter of 2018 through the third calendar quarter of 2021. We then get the year-on-year change in panel sales that corresponds to these data points (i.e., for the same restaurant brand, over the same period of time). Finally, we compute the correlation between the population-level sales measures and their panel-level counterparts separately for each firm.

Table W 3 shows for each of the 36 restaurant brands the parent company, the restaurant brand (when it is applicable), the ticker symbol of the parent company, the evaluation metric, and the correlation between the population-level and panel-level sales measure.

Table W 3: Company List with Panel to Population Correlation

| Company               | Sub-brand<br>(when applicable) | Ticker | Evaluation Metric       | Correlation |
|-----------------------|--------------------------------|--------|-------------------------|-------------|
| BJ's Restaurants      |                                | BJRI   | Sales                   | 98.9%       |
| Bloomin' Brands       |                                | BLMN   | Sales                   | 98.8%       |
| Brinker International | Chili's                        | EAT    | Same-store sales growth | 98.8%       |
| CEC Entertainment     |                                | CEC    | Store sales             | 64.3%       |
| Cheesecake Factory    |                                | CAKE   | Store sales             | 99.4%       |

|                          |             |      |                         |       |
|--------------------------|-------------|------|-------------------------|-------|
| Chipotle                 |             | CMG  | Sales                   | 73.5% |
| Chuy's                   |             | CHUY | Sales                   | 98.5% |
| Cracker Barrel           |             | CBRL | Sales                   | 99.7% |
| Darden                   |             | DRI  | Sales                   | 99.7% |
| Del Taco                 |             | TACO | Same-store sales growth | 53.0% |
| Denny's                  |             | DENN | Same-store sales growth | 98.6% |
| Dine Brands Global       | Applebee's  | DIN  | Same-store sales growth | 99.6% |
| Dine Brands Global       | IHOP        | DIN  | Same-store sales growth | 99.9% |
| Domino's Pizza           |             | DPZ  | Same-store sales growth | 91.6% |
| Dunkin' Brands           |             | DNKN | Sales                   | 93.0% |
| El Pollo Loco            |             | LOCO | Same-store sales growth | 61.9% |
| Fiesta Restaurant Group  |             | FRGI | Sales                   | 86.7% |
| Jack in the Box          |             | JACK | Same-store sales growth | 97.0% |
| Kura Sushi USA           |             | KRUS | Sales                   | 99.2% |
| Luby's                   |             | LUB  | Store sales             | 98.9% |
| McDonald's               |             | MCD  | Store sales             | 96.4% |
| Noodles                  |             | NDLS | Sales                   | 99.0% |
| Papa John's              |             | PZZA | Same-store sales growth | 98.8% |
| Potbelly                 |             | PBPB | Sales                   | 99.3% |
| Red Robin                |             | RRGB | Sales                   | 99.0% |
| Restaurant Brands        | Burger King | QSR  | Same-store sales growth | 88.7% |
| Restaurant Brands        | Popeyes     | QSR  | Same-store sales growth | 99.2% |
| Ruth's Hospitality Group |             | RUTH | Store sales             | 98.6% |
| Shake Shack              |             | SHAK | Sales                   | 97.9% |
| Starbucks                |             | SBUX | Store sales             | 99.1% |
| Texas Roadhouse          |             | TXRH | Store sales             | 98.1% |

|             |           |      |                         |       |
|-------------|-----------|------|-------------------------|-------|
| Wendy's     |           | WEN  | Same-store sales growth | 93.5% |
| Wingstop    |           | WING | Store sales             | 92.5% |
| Yum! Brands | KFC       | YUM  | Same-store sales growth | 73.6% |
| Yum! Brands | Pizza Hut | YUM  | Same-store sales growth | 64.3% |
| Yum! Brands | Taco Bell | YUM  | Same-store sales growth | 96.7% |

A simple average of the 36 restaurant-specific correlations is 91.8%. The median of these correlations is 98.6%, while the 25th and 75th percentiles are 92.3% and 99.1%, respectively.

### Web Appendix E: Matching Procedure

In our main analysis, we balance our panel across 98 cities, consisting of 49 treated cities which began e-scooter programs between June 2018 and October 2018, and 49 untreated cities which did not allow e-scooters to operate over the full duration of the observation period in two ways: (1) optimal matching at the city-level, and (2) coarsened exact matching (CEM) at the individual-level as follows.

First, we create a collection of the largest 100 candidate control cities by population size, which we confirmed did not allow e-scooter programs to operate over the observation period. We then pulled city-specific information from the U.S. Census Bureau and the National Oceanic and Atmospheric Administration to construct matching covariates including population, population density, average annual temperature, average annual precipitation, and median household income. We selected control cities using optimal full matching (Hansen and Klopfer, 2006). Table W 1 shows the resulting list of treatment and control cities.

We then apply CEM to match individuals from treated cities who rode an e-scooter at least one time with individuals in control cities. We create a set of matching covariates using transactional behavior over the pre-treatment period. Because our pre-treatment period is January 2017 to May 2018, however, we must do so in a parametrically parsimonious manner. We estimate individual-specific parameters representing the propensity to spend in each category through a random effects specification as follows:

$$(1) \quad \text{Spend}_{im(t),k} = \alpha_{0,k} + \beta_{0i,k} + \sum \alpha_{1m(t),k} \mathbb{I}\{m(t)\} + \epsilon_{im(t),k}$$

$$\beta_{0i,k} \sim N(0, \sigma_{\beta_{0,k}}^2), \epsilon_{im(t),k} \sim N(0, \sigma_{\epsilon,k}^2)$$

We assume that cross-sectional heterogeneity in the individual-specific spend parameters in category  $k$ ,  $\beta_{0i,k}$ , are governed by independent normal distributions, after controlling for city-specific shocks over time through city-specific intercept and month dummy variables  $\mathbb{I}\{m(t)\}$ . Our estimate of  $E[\beta_{0i,k} | \text{Spend}]$  summarizes the underlying baseline propensity of customer  $i$  to spend in category  $k$ . To model the propensity to be e-scooter riders using his-

torical consumption behavior, we use category-level spending in the following categories: Events & Attractions (e.g. AMC, Ticketmaster), Home Entertainment (e.g. Netflix, Nintendo), Restaurant (e.g. Panera Bread, Starbucks), Restaurant Delivery (e.g. DoorDash), Gas Station (e.g. Shell, Circle K), Mass Transit (e.g. Metrocard, Amtrak), Ridesharing (e.g. Uber, Lyft), and total spending as a proxy for income. We include these eight parameter estimates in our set of matching covariates. This significantly reduces the dimensionality of historical consumption behavior relative to matching directly upon weekly spending over the duration of the pre-treatment period, which would have left us with a significant number of unmatched treated individuals.

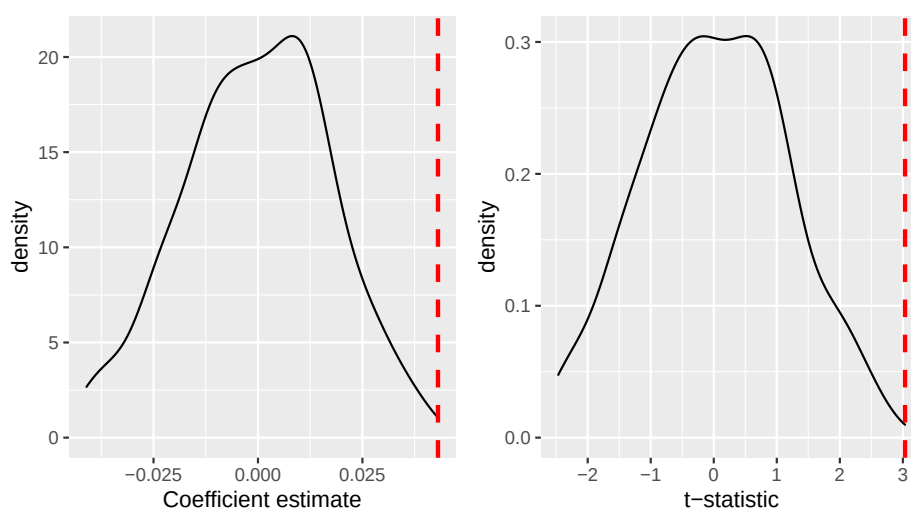
In summary, we find the “closest” control city to each treatment city based upon measurable city-specific statistics and then perform CEM in the usual way, assigning individuals to strata, pruning unmatched strata that have only treated individuals or only control individuals, and re-weighting individuals within each stratum so that treated and untreated individuals collectively have the same weight as one another. This procedure significantly reduced covariate imbalance for the empirical distribution of pre-period variables between the treated and control groups. This procedure left us with 13,583 treated individuals and 32,868 untreated individuals with strictly positive CEM weights, with 0.5% and 56.8% of treated and control units dropped due to the matching procedure.

### Web Appendix F: Permutation Test

As noted in the Empirical Results section of the main paper, we carry out a permutation test to evaluate whether our results are a byproduct of spurious common time trends. Each city is associated with a binary vector representing whether an e-scooter program had already been launched or not at each time point. We randomly permute this vector by city (e.g., randomly assigning Baltimore, Maryland's treatment timing to Arlington, Virginia), while leaving individuals' home city, restaurant spending, and other data unpermuted. We then re-run Equation 1 on the permuted data, obtain the coefficient estimates and corresponding t-statistic values associated with e-scooter program introduction,  $\beta_1$ , and repeat this procedure 200 times.

Figure W 1 presents the null distributions for the point estimate and t-statistic for e-scooter program on restaurant spending,  $\beta_1$  from Equation 1 in the main paper (in black). The point estimate and t-statistic for  $\beta_1$  using the original unpermuted data is shown in red. As noted in the main paper, the t-statistic for  $\beta_1$  using the unpermuted data is significantly different from the null distribution ( $p < .001$ ), implying that our original estimate is indeed significantly different from what we would have expected under the null hypothesis that there is no relationship between restaurant spending and e-scooter program introduction.

Figure W 1: Permutation Test for Restaurant Spending



*Note:* This figure shows the distribution of the estimates and t-statistics for the main coefficient of interest,  $\beta_1$  of Equation 1 in the main paper using the analysis performed by Christian and Barrett (2017) and Barron, Kung, and Proserpio (2021), as described in the Empirical Results section of the main paper. The red dashed lines represent the point estimate and t-statistic obtained using the original unpermuted data.

*Web Appendix G: Other Robustness Checks*

Table W 4: Robustness to Different City Pairing

|               | <i>Dependent variable:</i> |                         |                         |
|---------------|----------------------------|-------------------------|-------------------------|
|               | Total                      | Fast Food               | Sitdown                 |
|               | (1)                        | (2)                     | (3)                     |
| Scooter Entry | 0.06830***<br>(0.02126)    | 0.05589***<br>(0.01848) | 0.02014***<br>(0.00745) |
| Individual FE | ✓                          | ✓                       | ✓                       |
| Time FE       | ✓                          | ✓                       | ✓                       |
| $R^2$         | 0.227                      | 0.234                   | 0.076                   |
| No. of Obs.   | 4,681,875                  | 4,661,250               | 4,208,325               |

*Note:* Model setting is the same as that of our main analysis, except that we perturb the pairings of treatment and control cities using different city matching covariates. Instead of average annual temperature, we use average summer temperature and average winter temperature to better capture fluctuation in temperature over the year. We then performed CEM, as before, to match individuals within these cities. Dependent variable is total restaurant spending and each subcategory. Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table W 5: Other Robustness Checks

|               | <i>Dependent variable:</i> |                        |                         |                             |
|---------------|----------------------------|------------------------|-------------------------|-----------------------------|
|               | Quadratic Trend            | Daily Aggregated       | Monthly Aggregated      | One-period Treatment Window |
|               | (1)                        | (2)                    | (3)                     | (4)                         |
| Scooter Entry | 0.05919***<br>(0.01750)    | 0.01179**<br>(0.00482) | 0.08767***<br>(0.02946) | 0.04156**<br>(0.01732)      |
| Individual FE | ✓                          | ✓                      | ✓                       | ✓                           |
| Time FE       | ✓                          | ✓                      | ✓                       | ✓                           |
| $R^2$         | 0.338                      | 0.124                  | 0.467                   | 0.584                       |
| No. of Obs.   | 7,664,415                  | 53,650,905             | 1,765,138               | 324,681                     |

*Note:* Each model perturbs the main model specification (Equation 1 in the main paper) as follows: (1) city-specific quadratic trend is used instead of city-specific linear trend; (2) unit of time is one day instead of one week; (3) unit of time is one month instead of one week; (4) unit of time is five months, such that the treatment window (i.e., June to October 2018) represents a single time point. Robust standard errors (in parentheses) are clustered by city; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Analogous to the ‘daily aggregated’ and ‘monthly aggregated’ robustness checks shown in Table W 5, the ‘one-period treatment window’ result in Table W 5 aggregates the unit of

time to five months such that the treatment window (i.e., the window of time over which the cities we study introduce e-scooter programs, from June to October 2018) is collapsed into one unit of time. As a result, there is a pre-treatment period, one time point over which the e-scooter program introduction event occurs (conservatively excluded from the post-treatment period), and a post-treatment period.

### *Web Appendix H: Alternative Time Trend Specification*

As noted in Goodman-Bacon (2021), including city-specific time trends can bias our estimate of  $\beta$  in the presence of time-varying treatment effects or unobservables by underweighting time points at the very beginning and end of the dataset. To evaluate these concerns in our empirical setting, we run an alternative pre-treatment trend extrapolation specification. In this specification, we estimate  $\beta$  via a two-step procedure. In the first step, we estimate  $\tilde{\theta}_c$ , the coefficients associated with the time trend variables, off of the pre-treatment data. In the second step, we estimate  $\beta$ , the coefficient of interest, using the main model specification (Equation 1 in the main paper) conditional upon the estimated  $\tilde{\theta}_c$  variables.

We estimate the standard error associated with  $\beta$  via non-parametric bootstrap. We first sample with replacement from the original population of (treated and untreated) individuals in the credit card panel. We perform coarsened exact matching upon this bootstrapped population to create a matched set of treated individuals and estimate  $\tilde{\theta}_c(b)$ , the coefficients associated with the time trend variables estimated off of the pre-treatment data associated with the  $b$ th bootstrapped dataset. We then estimate  $\beta(b)$ , the point estimate for  $\beta$  associated with the  $b$ th bootstrapped dataset, as well as its corresponding standard error  $SE(\beta)(b)$ , conditional upon first-stage estimates  $\tilde{\theta}_c(b)$ . The unconditional standard error associated with  $\beta$  is equal to

$$SE(\beta) = \left( \text{Var}(E(\beta|\tilde{\theta}_c)) + E(\text{Var}(\beta|\tilde{\theta}_c)) \right)^{1/2}$$

where  $\text{Var}(E(\beta|\tilde{\theta}_c))$  is the empirical variance of the bootstrapped point estimates  $\beta(b)$  across the  $B$  bootstrap samples, and  $E(\text{Var}(\beta|\tilde{\theta}_c))$  is the empirical average of the squared standard errors associated with the bootstrapped estimates  $\beta(b)$ . We set the number of bootstraps  $B = 200$ .

The estimated treatment effect is positive and significant ( $\hat{\beta}$ : 0.0802, SE: .0355,  $p = .0239$ ), as in the main model specification.

*Web Appendix I: Number of Permitted E-scooters*

We collected data about the number of e-scooters allowed by local city governments in our treated cities, which we used to calculate spending lift per e-scooter allowed in the Policy Implications section of the main paper. However, this information is sometimes not always publicly disclosed. We were able to find data on the number of e-scooters allowed during the observation period for 32 of our 49 treated cities. All other cities are marked as ‘N/A’.

The total number of e-scooters allowed in each city is a function of the number of permitted e-scooter companies and the number of permitted devices per each operator. Usually both factors are dynamic. For example, Bolt, Lime, and Spin were the only companies allowed to deploy e-scooters in Portland in April 2019, but three additional companies were approved by August 2019. Erring on the side of conservatism, when more than one figure is available for e-scooters allowed in a given city, we report the maximum across all figures within the observation period as ‘E-scooters Allowed’ in Table W 6. Normalizing the number of e-scooters allowed by city population size, we also obtain the number of e-scooters allowed per 1,000 people to improve comparability. For our 32 cities, there were up to 58,380 e-scooters permitted in total, implying an average of 5.54 permitted e-scooters per 1,000 citizens. For city specific information, see Table W 6 below.

Table W 6: City List with Permitted E-scooter Fleet Size and That Per Capita

| City Name     | Scooters Allowed | Scooters Allowed /1K pop | Notes                                                                                                                                                                                          |
|---------------|------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abilene, TX   | N/A              | N/A                      | N/A                                                                                                                                                                                            |
| Ann Arbor, MI | N/A              | N/A                      | N/A                                                                                                                                                                                            |
| Arlington, VA | 2,600            | 11.14                    | A fleet cap of 350 devices per company was applied to six operators (Lime, Lyft, Spin, JUMP, Skip, and Bolt), while Bird was allowed to deploy up to 500 in June 2019. (Arlington County 2019) |
| Athens, GA    | N/A              | N/A                      | N/A                                                                                                                                                                                            |

|                  |       |      |                                                                                                                                             |
|------------------|-------|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Baltimore, MD    | 4,000 | 6.57 | Four operators (Bird, JUMP, Lime and Spin) were each allowed to deploy up to 1,000 e-scooters in March 2019. (City of Baltimore 2019)       |
| Bloomington, IN  | N/A   | N/A  | N/A                                                                                                                                         |
| Boise City, ID   | 750   | 3.32 | Bird, Lime, and Spin were allowed to deploy up to 750 e-scooters, collectively, in May 2019. (Fields 2019)                                  |
| Cary, NC         | N/A   | N/A  | N/A                                                                                                                                         |
| Cincinnati, OH   | 1,200 | 3.98 | Bird and Lime were allowed to deploy 1,200 e-scooters, collectively, in March 2019. (Wetterich 2019)                                        |
| Cleveland, OH    | 2,400 | 6.23 | Bird, Lime, Spin, and Veoride were allowed to deploy up to 2,400 devices, collectively, in July 2019. (City of Cleveland 2019)              |
| Columbia, MO     | 500   | 4.12 | Bird was allowed to deploy 500 e-scooters in November 2018. (City of Columbia 2018)                                                         |
| Columbus, OH     | 4,000 | 4.55 | Eight companies were permitted to deploy up to 500 e-scooters each as of summer 2019. (Gray 2022)                                           |
| Coral Gables, FL | 150   | 2.99 | Bird and Spin were allowed to deploy up to 150 e-scooters in total in October 2018. (City of Coral Gables 2018)                             |
| Dallas, TX       | 6,000 | 4.51 | Bird, Lime, and Razor were allowed to deploy up to 3,000, 2,000, and 1,000 e-scooters, respectively, in October 2018. (City of Dallas 2018) |
| Detroit, MI      | 1,200 | 1.78 | Three e-scooter operators (Bird, Lime, and Spin) were allowed to distribute up to 400 e-scooters each in July 2019. (Brown 2019)            |
| Durham, NC       | 800   | 2.97 | Bird, Gotcha, Lime, and Spin were allowed to deploy up to 200 e-scooters each in June 2019. (Daniels 2019)                                  |
| East Lansing, MI | N/A   | N/A  | N/A                                                                                                                                         |
| Fresno, CA       | N/A   | N/A  | N/A                                                                                                                                         |
| Greensboro, NC   | 400   | 1.37 | Bird and Lime were allowed to deploy up to 200 e-scooters each in January 2019. (City of Greensboro 2018)                                   |
| Indianapolis, IN | 6,000 | 6.94 | Six companies were permitted to deploy up to 1,000 e-scooters each in May 2019. (City of Indianapolis 2018)                                 |
| Kansas City, MO  | N/A   | N/A  | N/A                                                                                                                                         |

|                   |       |      |                                                                                                                                                        |
|-------------------|-------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| La Mesa, CA       | 300   | 5.04 | Bird was allowed to deploy up to 300 e-scooters in April 2019. (City of La Mesa 2018)                                                                  |
| Lansing, MI       | 500   | 4.27 | Lime and Bird were each allowed to deploy up to 250 e-scooters in October 2018. (Lehr 2018)                                                            |
| Long Beach, CA    | 1,800 | 3.86 | Six e-scooter operators including Bird, Lime, and Spin were allowed to deploy up to 300 e-scooters each in June 2018. (City of Long Beach 2019)        |
| Louisville, KY    | 1,200 | 1.94 | Bird, Lime, Bolt, and Spin were allowed to distribute up to 450, 450, 150, and 150 e-scooters, respectively, in August 2019. (City of Louisville 2019) |
| Lubbock, TX       | N/A   | N/A  | N/A                                                                                                                                                    |
| Milwaukee, WI     | 1,050 | 1.77 | Bird, Lime, and Spin were allowed to deploy up to 350 e-scooters each in August 2019. (City of Milwaukee 2019)                                         |
| Minneapolis, MN   | 2,000 | 4.76 | Four e-scooter operators (Lime, Lyft, Spin, and JUMP) were allowed to deploy up to 500 e-scooters each in October 2019. (City of Minneapolis 2019)     |
| Norman, OK        | 450   | 3.66 | Three e-scooter operators (Bird, Lime, and Veo) were each allowed to deploy up to 150 e-scooters each in January 2019. (City of Norman 2019)           |
| Oklahoma City, OK | 500   | 0.78 | Bird and Lime were allowed to deploy up to 250 e-scooters each in August 2018. (City of Oklahoma City 2018)                                            |
| Oxford, OH        | N/A   | N/A  | N/A                                                                                                                                                    |
| Peoria, IL        | N/A   | N/A  | N/A                                                                                                                                                    |
| Portland, OR      | 2,630 | 4.08 | Bolt, Lime, Spin, Razor, Shared and Bird were allowed to deploy 2,630 e-scooters, collectively, in August 2019. (City of Portland 2019)                |
| Providence, RI    | 600   | 3.34 | Spin and Veoride were each allowed to deploy up to 300 e-scooters in November 2019. (Machado 2019)                                                     |
| Raleigh, NC       | 1,500 | 3.23 | The maximum number of e-scooters that could be deployed across all operators, collectively, was 1,500. (City of Raleigh 2019)                          |
| Richmond, VA      | 1,000 | 4.41 | Bolt and Bird were allowed to deploy up to 500 e-scooters each in June 2019 and October 2019, respectively. (City of Richmond 2019)                    |
| Riverside, CA     | N/A   | N/A  | N/A                                                                                                                                                    |
| Russellville, AR  | N/A   | N/A  | N/A                                                                                                                                                    |

|                      |       |       |                                                                                                                                                  |
|----------------------|-------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Salt Lake City, UT   | 1,500 | 7.59  | Three e-scooter companies (Lime, Bird, and Spin) were each allowed to deploy up to 500 e-scooters in May 2019. (Raymond 2019)                    |
| San Antonio, TX      | 2,000 | 1.33  | Razor and Bird were allowed to deploy up to 2,000 e-scooters, collectively, in December 2019. (City of San Antonio 2019)                         |
| Santa Barbara, CA    | N/A   | N/A   | N/A                                                                                                                                              |
| Santa Clara, CA      | N/A   | N/A   | N/A                                                                                                                                              |
| Santa Cruz, CA       | N/A   | N/A   | N/A                                                                                                                                              |
| South Lake Tahoe, CA | 550   | 25.07 | Lime was allowed to deploy up to 550 e-scooters in April 2019. (City of South Lake Tahoe 2019)                                                   |
| St. Louis, MO        | 7,500 | 24.34 | Three e-scooter companies (Bird, Lime, and Spin) were each allowed to deploy up to 2,500 e-scooters in May 2019. (City of St. Louis 2019)        |
| St. Paul, MN         | 2,000 | 6.57  | Lime, Bird, and Spin were allowed to deploy up to 2,000 e-scooters, collectively, as part of the city's 2019 e-scooter program. (Zbikowski 2019) |
| Statesboro, GA       | N/A   | N/A   | N/A                                                                                                                                              |
| Stillwater, OK       | 300   | 6.01  | Spin was allowed to deploy up to 300 e-scooters in January 2019. (City of Stillwater 2019)                                                       |
| Tacoma, WA           | 1,000 | 4.70  | Lime and Bird were each allowed to operate up to 500 e-scooters in January 2019. (City of Tacoma 2018)                                           |

*Web Appendix J: Calculation of Incremental Profit Lift in Dallas*

In this section, we provide additional detail regarding the calculation of incremental contribution profit to the restaurant sector, as discussed in the Policy Implications section of the main paper. Our analysis proceeds as follows:

1. Estimated incremental restaurant spending in Dallas is \$423 thousand.
2. We estimate the contribution margin for the restaurant sector through an analysis of publicly-disclosed financial data disclosed in annual reports filed with the Securities and Exchange Commission (SEC) in 2018 for the largest publicly-traded restaurant companies by market capitalization including Cheesecake Factory, Chipotle, Darden Restaurants (owner of Olive Garden, LongHorn Steakhouse, Cheddar's Scratch Kitchen, Yard House, The Capital Grille, Seasons 52, Bahama Breeze, and Eddie V's), Jamba Juice, Krispy Kreme, McDonald's, Restaurant Brands International (owner of Burger King, Popeyes Louisiana Kitchen, and Tim Hortons), Starbucks, Texas Roadhouse, Wingstop, Yum Brands (owner of KFC, Pizza Hut, and Taco Bell).
3. For each company, we estimate its contribution profit margin as a percentage of sales in 2018 by obtaining total revenue for owned stores only (i.e., excluding franchised stores and other revenue sources), and deducting all direct expenses (e.g., cost of food, labor, rent, and other direct operating expenses).<sup>1</sup> We then take a simple average of the resulting contribution margins across all companies. This results in a contribution margin percentage of 22.3% for the restaurant category.
4. To arrive at total contribution profit lift, we multiply estimated sales lift figure by our estimate of contribution margin. The resulting total contribution profit lift across restaurant brands in Dallas is \$94,380.

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